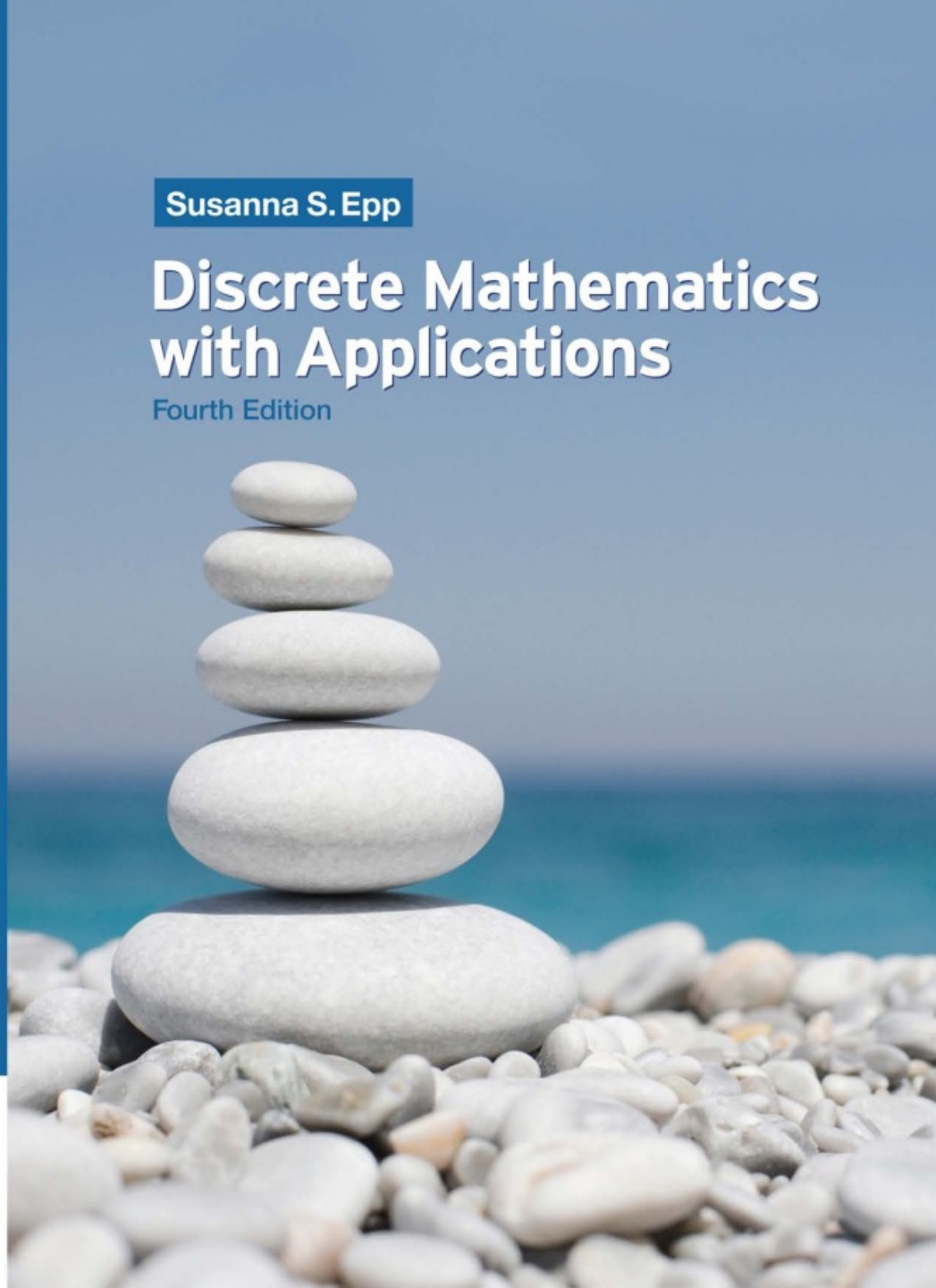




Susanna S. Epp

Discrete Mathematics with Applications

Fourth Edition

Exercises of Discrete Structures

*You may need to refer to the textbook to get the context of these exercises.



Exercise Set 2.1

Write truth tables for the statement forms in 12–15.

12. $\sim p \wedge q$

13. $\sim(p \wedge q) \vee (p \vee q)$

14. $p \wedge (q \wedge r)$

15. $p \wedge (\sim q \vee r)$

Determine whether the statement forms in 16–24 are logically equivalent. In each case, construct a truth table and include a sentence justifying your answer. Your sentence should show that you understand the meaning of logical equivalence.

16. $p \vee (p \wedge q)$ and p

17. $\sim(p \wedge q)$ and $\sim p \wedge \sim q$

18. $p \vee \mathbf{t}$ and $\mathbf{t}$

19. $p \wedge \mathbf{t}$ and p

20. $p \wedge \mathbf{c}$ and $p \vee \mathbf{c}$

21. $(p \wedge q) \wedge r$ and $p \wedge (q \wedge r)$

22. $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$

23. $(p \wedge q) \vee r$ and $p \wedge (q \vee r)$

24. $(p \vee q) \vee (p \wedge r)$ and $(p \vee q) \wedge r$



Use truth tables to establish which of the statement forms in 40–43 are tautologies and which are contradictions.

40. $(p \wedge q) \vee (\sim p \vee (p \wedge \sim q))$

41. $(p \wedge \sim q) \wedge (\sim p \vee q)$

42. $((\sim p \wedge q) \wedge (q \wedge r)) \wedge \sim q$

43. $(\sim p \vee q) \vee (p \wedge \sim q)$

Use Theorem 2.1.1 to verify the logical equivalences in 50–54.
Supply a reason for each step.

$$\mathbf{50.} \quad (p \wedge \sim q) \vee p \equiv p \qquad \mathbf{51.} \quad p \wedge (\sim q \vee p) \equiv p$$

$$\mathbf{52.} \quad \sim(p \vee \sim q) \vee (\sim p \wedge \sim q) \equiv \sim p$$

$$\mathbf{53.} \quad \sim((\sim p \wedge q) \vee (\sim p \wedge \sim q)) \vee (p \wedge q) \equiv p$$

$$\mathbf{54.} \quad (p \wedge (\sim(\sim p \vee q))) \vee (p \wedge q) \equiv p$$



Exercise Set 2.2

Construct truth tables for the statement forms in 5–11.

5. $\sim p \vee q \rightarrow \sim q$

6. $(p \vee q) \vee (\sim p \wedge q) \rightarrow q$

7. $p \wedge \sim q \rightarrow r$

8. $\sim p \vee q \rightarrow r$

9. $p \wedge \sim r \leftrightarrow q \vee r$

10. $(p \rightarrow r) \leftrightarrow (q \rightarrow r)$

11. $(p \rightarrow (q \rightarrow r)) \leftrightarrow ((p \wedge q) \rightarrow r)$

15. Determine whether the following statement forms are logically equivalent:

$$p \rightarrow (q \rightarrow r) \quad \text{and} \quad (p \rightarrow q) \rightarrow r$$

In 16 and 17, write each of the two statements in symbolic form and determine whether they are logically equivalent. Include a truth table and a few words of explanation.

- 16.** If you paid full price, you didn't buy it at Crown Books.
You didn't buy it at Crown Books or you paid full price.
17. If 2 is a factor of n and 3 is a factor of n , then 6 is a factor of n . 2 is not a factor of n or 3 is not a factor of n or 6 is a factor of n .

If statement forms P and Q are logically equivalent, then $P \leftrightarrow Q$ is a tautology. Conversely, if $P \leftrightarrow Q$ is a tautology, then P and Q are logically equivalent. Use $\leftrightarrow$ to convert each of the logical equivalences in 29–31 to a tautology. Then use a truth table to verify each tautology.

29. $p \rightarrow (q \vee r) \equiv (p \wedge \sim q) \rightarrow r$

30. $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

31. $p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Rewrite the statements in 34 and 35 in if-then form in two ways, one of which is the contrapositive of the other.

- 34.** The Cubs will win the pennant only if they win tomorrow's game.
35. Sam will be allowed on Signe's racing boat only if he is an expert sailor.

In 37–39, rewrite the statements in if-then form.

37. Payment will be made on the fifth unless a new hearing is granted.

38. Ann will go unless it rains.

39. This door will not open unless a security code is entered.

Rewrite the statements in 40 and 41 in if-then form.

40. Catching the 8:05 bus is a sufficient condition for my being on time for work.

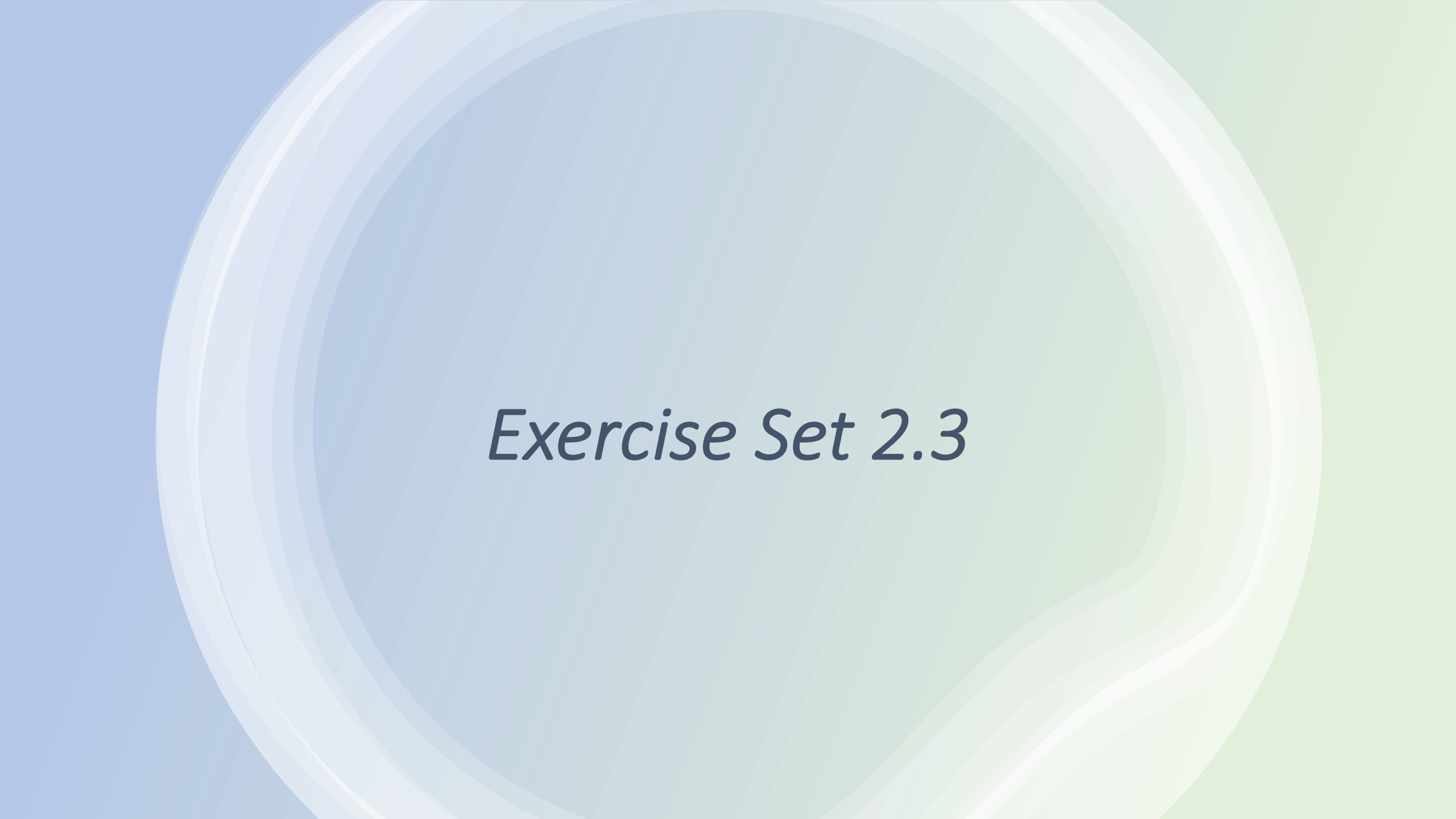
41. Having two 45° angles is a sufficient condition for this triangle to be a right triangle.

Use the contrapositive to rewrite the statements in 42 and 43 in if-then form in two ways.

- 42. Being divisible by 3 is a necessary condition for this number to be divisible by 9.
- 43. Doing homework regularly is a necessary condition for Jim to pass the course.

Note that “a sufficient condition for s is r ” means r is a sufficient condition for s and that “a necessary condition for s is r ” means r is a necessary condition for s . Rewrite the statements in 44 and 45 in if-then form.

- 44. A sufficient condition for Jon’s team to win the championship is that it win the rest of its games.
- 45. A necessary condition for this computer program to be correct is that it not produce error messages during translation.



Exercise Set 2.3

Use truth tables to determine whether the argument forms in 6–11 are valid. Indicate which columns represent the premises and which represent the conclusion, and include a sentence explaining how the truth table supports your answer. Your explanation should show that you understand what it means for a form of argument to be valid or invalid.

$$\begin{array}{l} 6. \quad p \rightarrow q \\ \quad q \rightarrow p \\ \therefore p \vee q \end{array}$$

$$\begin{array}{l} 7. \quad p \\ \quad p \rightarrow q \\ \quad \sim q \vee r \\ \therefore r \end{array}$$

$$\begin{array}{l} 8. \quad p \vee q \\ \quad p \rightarrow \sim q \\ \quad p \rightarrow r \\ \therefore r \end{array}$$

$$\begin{array}{l} 9. \quad p \wedge q \rightarrow \sim r \\ \quad p \vee \sim q \\ \quad \sim q \rightarrow p \\ \therefore \sim r \end{array}$$

$$10. \quad p \rightarrow r$$

$$q \rightarrow r$$

$$\therefore p \vee q \rightarrow r$$

$$11. \quad p \rightarrow q \vee r$$

$$\sim q \vee \sim r$$

$$\therefore \sim p \vee \sim r$$

12. Use truth tables to show that the following forms of argument are invalid.

a. $p \rightarrow q$

$$q$$

$$\therefore p$$

(converse error)

b. $p \rightarrow q$

$$\sim p$$

$$\therefore \sim q$$

(inverse error)

37. In the back of an old cupboard you discover a note signed by a pirate famous for his bizarre sense of humor and love of logical puzzles. In the note he wrote that he had hidden treasure somewhere on the property. He listed five true statements (a–e below) and challenged the reader to use them to figure out the location of the treasure.

- a. If this house is next to a lake, then the treasure is not in the kitchen.
- b. If the tree in the front yard is an elm, then the treasure is in the kitchen.
- c. This house is next to a lake.
- d. The tree in the front yard is an elm or the treasure is buried under the flagpole.
- e. If the tree in the back yard is an oak, then the treasure is in the garage.

Where is the treasure hidden?

39. The famous detective Percule Hoirot was called in to solve a baffling murder mystery. He determined the following facts:

- a. Lord Hazelton, the murdered man, was killed by a blow on the head with a brass candlestick.
- b. Either Lady Hazelton or a maid, Sara, was in the dining room at the time of the murder.

- c. If the cook was in the kitchen at the time of the murder, then the butler killed Lord Hazelton with a fatal dose of strychnine.
- d. If Lady Hazelton was in the dining room at the time of the murder, then the chauffeur killed Lord Hazelton.
- e. If the cook was not in the kitchen at the time of the murder, then Sara was not in the dining room when the murder was committed.
- f. If Sara was in the dining room at the time the murder was committed, then the wine steward killed Lord Hazelton.

Is it possible for the detective to deduce the identity of the murderer from these facts? If so, who did murder Lord Hazelton? (Assume there was only one cause of death.)

40. Sharky, a leader of the underworld, was killed by one of his own band of four henchmen. Detective Sharp interviewed the men and determined that all were lying except for one. He deduced who killed Sharky on the basis of the following statements:

- a. Socko: Lefty killed Sharky.
- b. Fats: Muscles didn't kill Sharky.
- c. Lefty: Muscles was shooting craps with Socko when Sharky was knocked off.
- d. Muscles: Lefty didn't kill Sharky.

Who did kill Sharky?

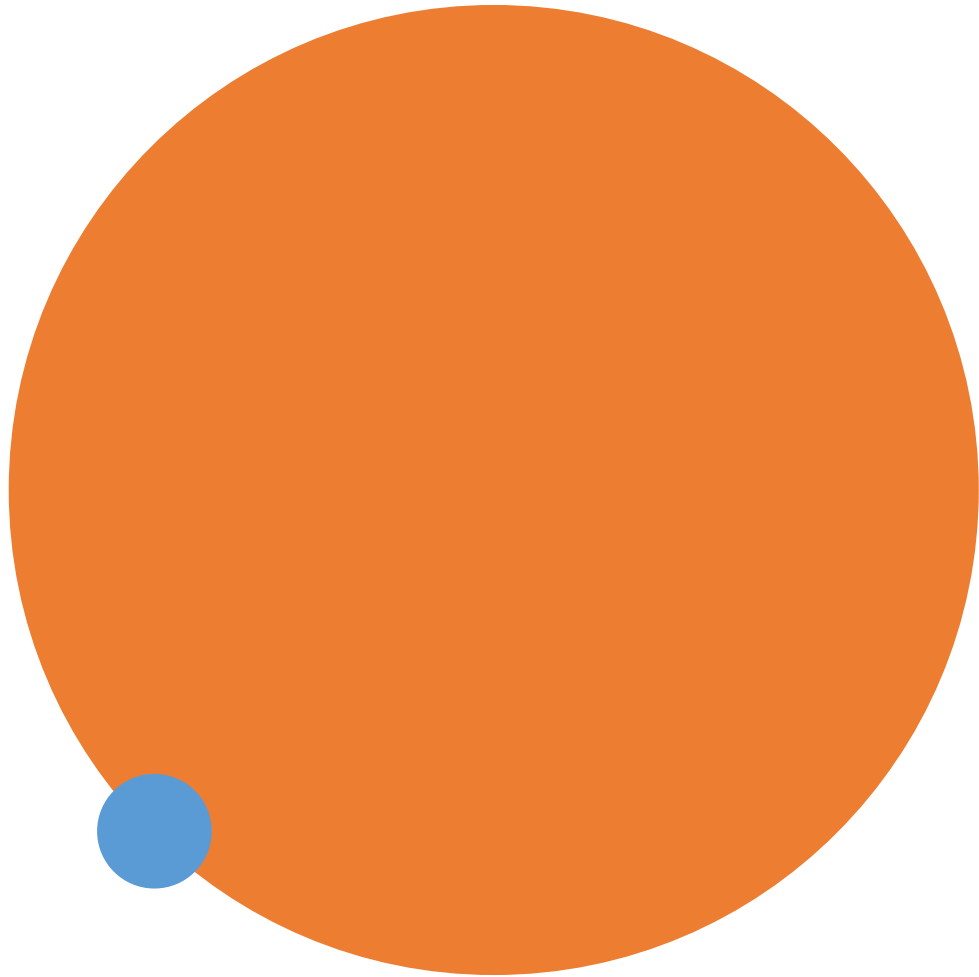
In 41–44 a set of premises and a conclusion are given. Use the valid argument forms listed in Table 2.3.1 to deduce the conclusion from the premises, giving a reason for each step as in Example 2.3.8. Assume all variables are statement variables.

41. a. $\sim p \vee q \rightarrow r$
b. $s \vee \sim q$
c. $\sim t$
d. $p \rightarrow t$
e. $\sim p \wedge r \rightarrow \sim s$
f. $\therefore \sim q$

43. a. $\sim p \rightarrow r \wedge \sim s$
b. $t \rightarrow s$
c. $u \rightarrow \sim p$
d. $\sim w$
e. $u \vee w$
f. $\therefore \sim t$

42. a. $p \vee q$
b. $q \rightarrow r$
c. $p \wedge s \rightarrow t$
d. $\sim r$
e. $\sim q \rightarrow u \wedge s$
f. $\therefore t$

44. a. $p \rightarrow q$
b. $r \vee s$
c. $\sim s \rightarrow \sim t$
d. $\sim q \vee s$
e. $\sim s$
f. $\sim p \wedge r \rightarrow u$
g. $w \vee t$
h. $\therefore u \wedge w$



- Give 3 examples for each type of fallacy in chapter 1.





Exercise Set 3.1

2. Indicate which of the following statements are true and which are false. Justify your answers as best as you can.
- a. Every integer is a real number.
 - b. 0 is a positive real number.
 - c. For all real numbers r , $-r$ is a negative real number.
 - d. Every real number is an integer.
3. Let $P(x)$ be the predicate " $x > 1/x$."
- a. Write $P(2)$, $P(\frac{1}{2})$, $P(-1)$, $P(-\frac{1}{2})$, and $P(-8)$, and indicate which of these statements are true and which are false.
 - b. Find the truth set of $P(x)$ if the domain of x is $\mathbf{R}$, the set of all real numbers.
 - c. If the domain is the set $\mathbf{R}^+$ of all positive real numbers, what is the truth set of $P(x)$?

4. Let $Q(n)$ be the predicate “ $n^2 \leq 30$.”
- a. Write $Q(2)$, $Q(-2)$, $Q(7)$, and $Q(-7)$, and indicate which of these statements are true and which are false.
 - b. Find the truth set of $Q(n)$ if the domain of n is $\mathbf{Z}$, the set of all integers.
 - c. If the domain is the set $\mathbf{Z}^+$ of all positive integers, what is the truth set of $Q(n)$?
5. Let $Q(x, y)$ be the predicate “If $x < y$ then $x^2 < y^2$ ” with domain for both x and y being the set $\mathbf{R}$ of real numbers.
- a. Explain why $Q(x, y)$ is false if $x = -2$ and $y = 1$.
 - b. Give values different from those in part (a) for which $Q(x, y)$ is false.
 - c. Explain why $Q(x, y)$ is true if $x = 3$ and $y = 8$.
 - d. Give values different from those in part (c) for which $Q(x, y)$ is true.

Find counterexamples to show that the statements in 9–12 are false.

9. $\forall x \in \mathbf{R}, x > 1/x.$

10. $\forall a \in \mathbf{Z}, (a - 1)/a$ is not an integer.

11. $\forall$ positive integers m and n , $m \cdot n \geq m + n.$

12. $\forall$ real numbers x and y , $\sqrt{x + y} = \sqrt{x} + \sqrt{y}.$

16. Rewrite each of the following statements in the form

“ $\forall$ _____ x , _____.”

- a. All dinosaurs are extinct.
- b. Every real number is positive, negative, or zero.
- c. No irrational numbers are integers.
- d. No logicians are lazy.
- e. The number 2,147,581,953 is not equal to the square of any integer.
- f. The number -1 is not equal to the square of any real number.

17. Rewrite each of the following in the form “ $\exists$ _____ x such that _____.”

- a. Some exercises have answers.
- b. Some real numbers are rational.

18. Let D be the set of all students at your school, and let $M(s)$ be “ s is a math major,” let $C(s)$ be “ s is a computer science student,” and let $E(s)$ be “ s is an engineering student.” Express each of the following statements using quantifiers, variables, and the predicates $M(s)$, $C(s)$, and $E(s)$.
- a. There is an engineering student who is a math major.
 - b. Every computer science student is an engineering student.
 - c. No computer science students are engineering students.
 - d. Some computer science students are also math majors.
 - e. Some computer science students are engineering students and some are not.

19. Consider the following statement:

$\forall$ integers n , if n^2 is even then n is even.

Which of the following are equivalent ways of expressing this statement?

- a. All integers have even squares and are even.
- b. Given any integer whose square is even, that integer is itself even.
- c. For all integers, there are some whose square is even.
- d. Any integer with an even square is even.
- e. If the square of an integer is even, then that integer is even.
- f. All even integers have even squares.

22. Rewrite each of the following statements in the form “ $\forall$ _____ x , if _____ then _____.”

- a. All Java programs have at least 5 lines.
- b. Any valid argument with true premises has a true conclusion.

23. Rewrite each of the following statements in the two forms “ $\forall x$, if _____ then _____” and “ $\forall$ _____ x , _____” (without an if-then).

- a. All equilateral triangles are isosceles.
- b. Every computer science student needs to take data structures.

24. Rewrite the following statements in the two forms “ $\exists$ _____ x such that _____” and “ $\exists x$ such that _____ and _____.”

- a. Some hatters are mad.
- b. Some questions are easy.

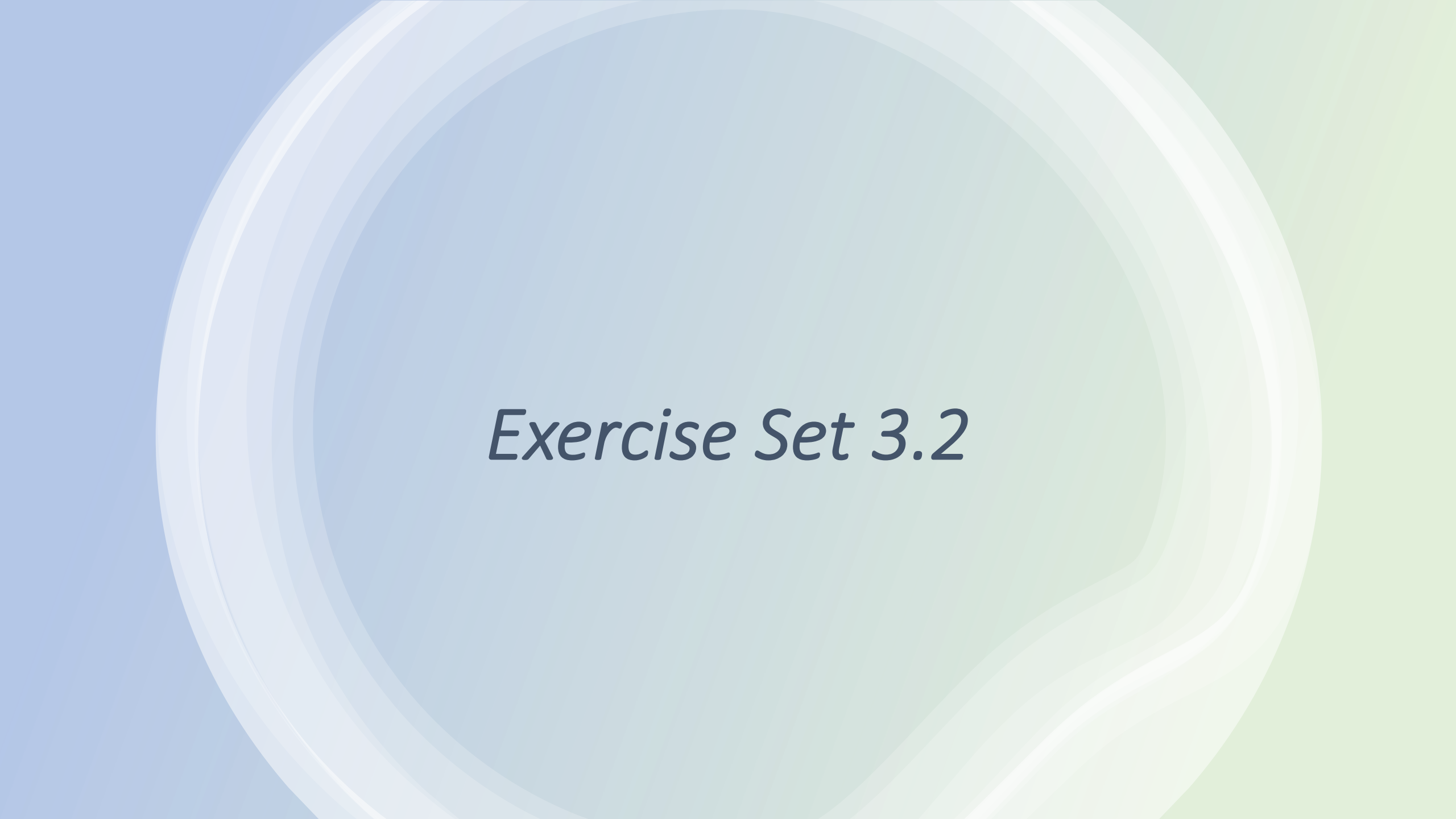
25. The statement “The square of any rational number is rational” can be rewritten formally as “For all rational numbers x , x^2 is rational” or as “For all x , if x is rational then x^2 is rational.” Rewrite each of the following statements in the two forms “ $\forall$ _____ x , _____” and “ $\forall x$, if _____, then _____” or in the two forms “ $\forall$ _____ x and y , _____” and “ $\forall x$ and y , if _____, then _____.”

- a. The reciprocal of any nonzero fraction is a fraction.
- b. The derivative of any polynomial function is a polynomial function.
- c. The sum of the angles of any triangle is 180° .
- d. The negative of any irrational number is irrational.
- e. The sum of any two even integers is even.
- f. The product of any two fractions is a fraction.

In 28–30, rewrite each statement without using quantifiers or variables. Indicate which are true and which are false, and justify your answers as best as you can.

28. Let the domain of x be the set D of objects discussed in mathematics courses, and let $\text{Real}(x)$ be “ x is a real number,” $\text{Pos}(x)$ be “ x is a positive real number,” $\text{Neg}(x)$ be “ x is a negative real number,” and $\text{Int}(x)$ be “ x is an integer.”
- a. $\text{Pos}(0)$
 - b. $\forall x, \text{Real}(x) \wedge \text{Neg}(x) \rightarrow \text{Pos}(-x).$
 - c. $\forall x, \text{Int}(x) \rightarrow \text{Real}(x).$
 - d. $\exists x$ such that $\text{Real}(x) \wedge \sim \text{Int}(x).$
29. Let the domain of x be the set of geometric figures in the plane, and let $\text{Square}(x)$ be “ x is a square” and $\text{Rect}(x)$ be “ x is a rectangle.”
- a. $\exists x$ such that $\text{Rect}(x) \wedge \text{Square}(x).$
 - b. $\exists x$ such that $\text{Rect}(x) \wedge \sim \text{Square}(x).$
 - c. $\forall x, \text{Square}(x) \rightarrow \text{Rect}(x).$

30. Let the domain of x be the set $\mathbf{Z}$ of integers, and let $\text{Odd}(x)$ be “ x is odd,” $\text{Prime}(x)$ be “ x is prime,” and $\text{Square}(x)$ be “ x is a perfect square.” (An integer n is said to be a **perfect square** if, and only if, it equals the square of some integer. For example, 25 is a perfect square because $25 = 5^2$.)
- a. $\exists x$ such that $\text{Prime}(x) \wedge \sim \text{Odd}(x)$.
 - b.** $\forall x, \text{Prime}(x) \rightarrow \sim \text{Square}(x)$.
 - c. $\exists x$ such that $\text{Odd}(x) \wedge \text{Square}(x)$.



Exercise Set 3.2

1. Which of the following is a negation for “All discrete mathematics students are athletic”? More than one answer may be correct.
- a. There is a discrete mathematics student who is nonathletic.
 - b. All discrete mathematics students are nonathletic.
 - c. There is an athletic person who is a discrete mathematics student.
 - d. No discrete mathematics students are athletic.
 - e. Some discrete mathematics students are nonathletic.
 - f. No athletic people are discrete mathematics students.

2. Which of the following is a negation for “All dogs are loyal”? More than one answer may be correct.
- a. All dogs are disloyal.
 - b. No dogs are loyal.
 - c. Some dogs are disloyal.
 - d. Some dogs are loyal.
 - e. There is a disloyal animal that is not a dog.
 - f. There is a dog that is disloyal.
 - g. No animals that are not dogs are loyal.
 - h. Some animals that are not dogs are loyal.
3. Write a formal negation for each of the following statements:
- a. $\forall$ fish x , x has gills.
 - b. $\forall$ computers c , c has a CPU.
 - c. $\exists$ a movie m such that m is over 6 hours long.
 - d. $\exists$ a band b such that b has won at least 10 Grammy awards.

In each of 11–14 determine whether the proposed negation is correct. If it is not, write a correct negation.

11. *Statement:* The sum of any two irrational numbers is irrational.

Proposed negation: The sum of any two irrational numbers is rational.

12. *Statement:* The product of any irrational number and any rational number is irrational.

Proposed negation: The product of any irrational number and any rational number is rational.

13. *Statement:* For all integers n , if n^2 is even then n is even.

Proposed negation: For all integers n , if n^2 is even then n is not even.

14. *Statement:* For all real numbers x_1 and x_2 , if $x_1^2 = x_2^2$ then $x_1 = x_2$.

Proposed negation: For all real numbers x_1 and x_2 , if $x_1^2 = x_2^2$ then $x_1 \neq x_2$.

In 16–23, write a negation for each statement.

16. $\forall$ real numbers x , if $x^2 \geq 1$ then $x > 0$.

17. $\forall$ integers d , if $6/d$ is an integer then $d = 3$.

18. $\forall x \in \mathbf{R}$, if $x(x + 1) > 0$ then $x > 0$ or $x < -1$.

19. $\forall n \in \mathbf{Z}$, if n is prime then n is odd or $n = 2$.

20. $\forall$ integers a , b and c , if $a - b$ is even and $b - c$ is even, then $a - c$ is even.

21. $\forall$ integers n , if n is divisible by 6, then n is divisible by 2 and n is divisible by 3.

22. If the square of an integer is odd, then the integer is odd.

23. If a function is differentiable then it is continuous.

In 26–33, for each statement in the referenced exercise write the converse, inverse, and contrapositive. Indicate as best as you can which among the statement, its converse, its inverse, and its contrapositive are true and which are false. Give a counterexample for each that is false.

26. Exercise 16

27. Exercise 17

28. Exercise 18

29. Exercise 19

30. Exercise 20

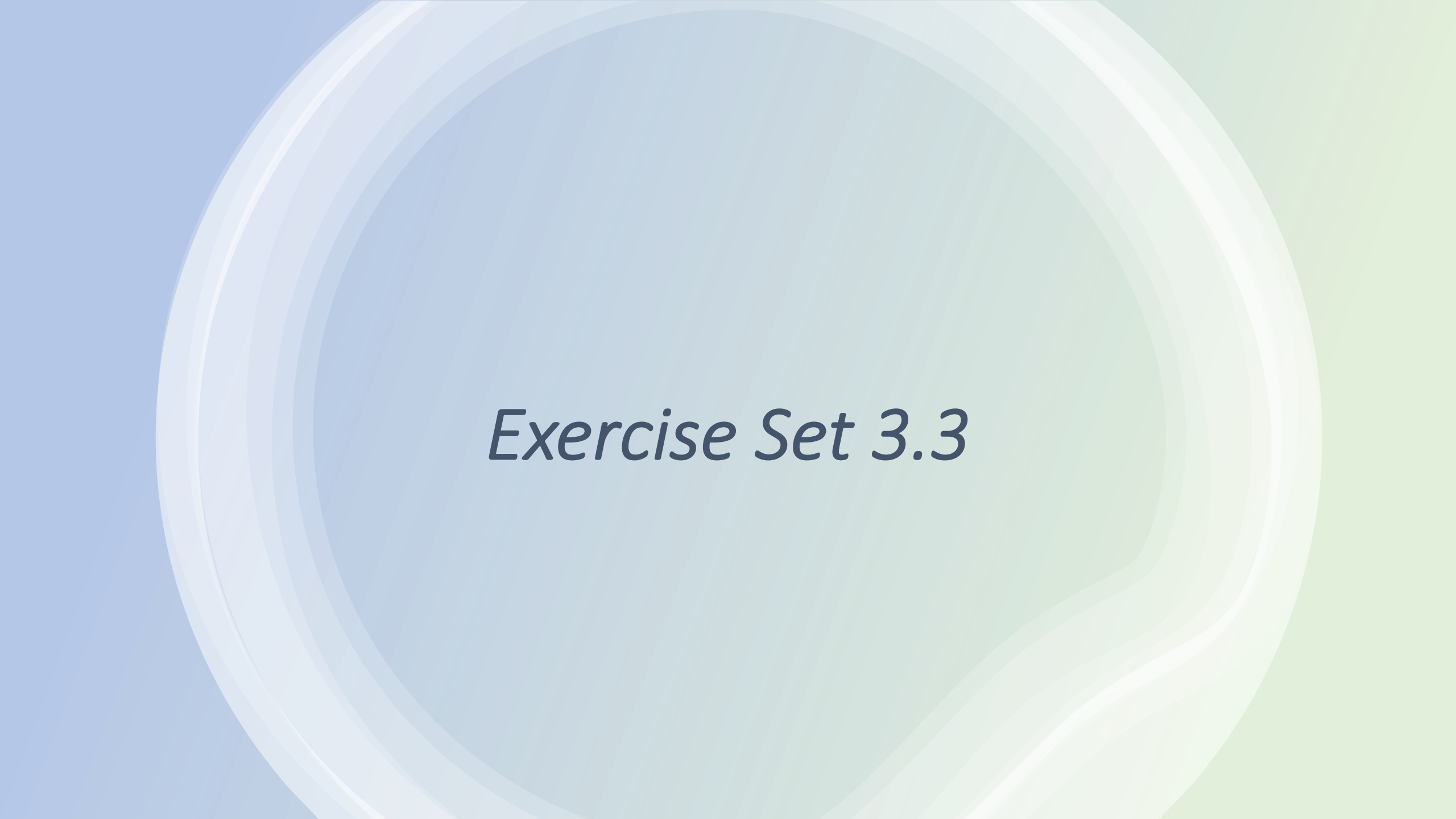
31. Exercise 21

32. Exercise 22

33. Exercise 23

Use the facts that the negation of a $\forall$ statement is a $\exists$ statement and that the negation of an if-then statement is an *and* statement to rewrite each of the statements 43–46 without using the word *necessary* or *sufficient*.

- 43. Being divisible by 8 is not a necessary condition for being divisible by 4.
- 44. Having a large income is not a necessary condition for a person to be happy.
- 45. Having a large income is not a sufficient condition for a person to be happy.
- 46. Being a polynomial is not a sufficient condition for a function to have a real root.



Exercise Set 3.3

4. The following statement is true: “ $\forall$ real numbers x , $\exists$ an integer n such that $n > x$.”* For each x given below, find an n to make the predicate “ $n > x$ ” true.
- a. $x = 15.83$ b. $x = 10^8$ c. $x = 10^{10^{10}}$

The statements in exercises 5–8 refer to the Tarski world given in Example 3.3.1. Explain why each is true.

5. For all circles x there is a square y such that x and y have the same color.
6. For all squares x there is a circle y such that x and y have different colors and y is above x .
7. There is a triangle x such that for all squares y , x is above y .
8. There is a triangle x such that for all circles y , y is above x .

9. Let $D = E = \{-2, -1, 0, 1, 2\}$. Explain why the following statements are true.
- a. $\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } x + y = 0.$
 - b. $\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, x + y = y.$
10. This exercise refers to Example 3.3.3. Determine whether each of the following statements is true or false.
- a. $\forall \text{ students } S, \exists \text{ a dessert } D \text{ such that } S \text{ chose } D.$
 - b. $\forall \text{ students } S, \exists \text{ a salad } T \text{ such that } S \text{ chose } T.$
 - c. $\exists \text{ a dessert } D \text{ such that } \forall \text{ students } S, S \text{ chose } D.$
 - d. $\exists \text{ a beverage } B \text{ such that } \forall \text{ students } D, D \text{ chose } B.$
 - e. $\exists \text{ an item } I \text{ such that } \forall \text{ students } S, S \text{ did not choose } I.$
 - f. $\exists \text{ a station } Z \text{ such that } \forall \text{ students } S, \exists \text{ an item } I \text{ such that } S \text{ chose } I \text{ from } Z.$

12. Let $D = E = \{-2, -1, 0, 1, 2\}$. Write negations for each of the following statements and determine which is true, the given statement or its negation.

- a. $\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } x + y = 1.$
- b. $\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, x + y = -y.$
- c. $\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } xy \geq y.$
- d. $\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, x \leq y.$

In each of 13–19, (a) rewrite the statement in English without using the symbol $\forall$ or $\exists$ or variables and expressing your answer as simply as possible, and (b) write a negation for the statement.

13. $\forall$ colors C , $\exists$ an animal A such that A is colored C .

14. $\exists$ a book b such that $\forall$ people p , p has read b .

15. $\forall$ odd integers n , $\exists$ an integer k such that $n = 2k + 1$.

16. $\exists$ a real number u such that $\forall$ real numbers v , $uv = v$.

17. $\forall r \in \mathbf{Q}$, $\exists$ integers a and b such that $r = a/b$.

18. $\forall x \in \mathbf{R}$, $\exists$ a real number y such that $x + y = 0$.

19. $\exists x \in \mathbf{R}$ such that for all real numbers y , $x + y = 0$.

In 22 and 23, rewrite each statement without using variables or the symbol $\forall$ or $\exists$. Indicate whether the statement is true or false.

22. a. $\forall$ real numbers x , $\exists$ a real number y such that $x + y = 0$.
b. $\exists$ a real number y such that $\forall$ real numbers x , $x + y = 0$.
23. a. $\forall$ nonzero real numbers r , $\exists$ a real number s such that $rs = 1$.
b. $\exists$ a real number r such that $\forall$ nonzero real numbers s , $rs = 1$.

Each statement in 25–28 refers to the Tarski world of Figure 3.3.1. For each, (a) determine whether the statement is true or false and justify your answer, (b) write a negation for the statement (referring, if you wish, to the result in exercise 24).

25. $\forall$ circles x and $\forall$ squares y , x is above y .

26. $\forall$ circles x and $\forall$ triangles y , x is above y .

27. $\exists$ a circle x and $\exists$ a square y such that x is above y and x and y have different colors.

28. $\exists$ a triangle x and $\exists$ a square y such that x is above y and x and y have the same color.

In 33–39, (a) rewrite the statement formally using quantifiers and variables, and (b) write a negation for the statement.

33. Everybody loves somebody.

34. Somebody loves everybody.

35. Everybody trusts somebody.

36. Somebody trusts everybody.

37. Any even integer equals twice some integer.

38. Every action has an equal and opposite reaction.

39. There is a program that gives the correct answer to every question that is posed to it.

In 46–54, refer to the Tarski world given in Figure 3.1.1, which is printed again here for reference. The domains of all variables consist of all the objects in the Tarski world. For each statement, (a) indicate whether the statement is true or false and justify your answer, (b) write the given statement using the formal logical notation illustrated in Example 3.3.10, and (c) write the negation of the given statement using the formal logical notation of Example 3.3.10.

 <i>a</i>	 <i>b</i>			
		 <i>c</i>	 <i>d</i>	
	 <i>e</i>	 <i>f</i>		
	 <i>g</i>	 <i>h</i>	 <i>i</i>	
			 <i>j</i>	 <i>k</i>

46. There is a triangle x such that for all squares y , x is above y .
47. There is a triangle x such that for all circles y , x is above y .
48. For all circles x , there is a square y such that y is to the right of x .
49. For every object x , there is an object y such that $x \neq y$ and x and y have different colors.
50. For every object x , there is an object y such that if $x \neq y$ then x and y have different colors.
51. There is an object y such that for all objects x , if $x \neq y$ then x and y have different colors.
52. For all circles x and for all triangles y , x is to the right of y .
53. There is a circle x and there is a square y such that x and y have the same color.
54. There is a circle x and there is a triangle y such that x and y have the same color.

Exercise Set 3.4

Use universal instantiation or universal modus ponens to fill in valid conclusions for the arguments in 2–4.

2. If an integer n equals $2 \cdot k$ and k is an integer, then n is even.

0 equals $2 \cdot 0$ and 0 is an integer.

$\therefore$ _____.

3. For all real numbers a , b , c , and d , if $b \neq 0$ and $d \neq 0$, then $a/b + c/d = (ad + bc)/bd$.

$a = 2$, $b = 3$, $c = 4$, and $d = 5$ are particular real numbers such that $b \neq 0$ and $d \neq 0$.

$\therefore$ _____.

Some of the arguments in 7–18 are valid by universal modus ponens or universal modus tollens; others are invalid and exhibit the converse or the inverse error. State which are valid and which are invalid. Justify your answers.

7. All healthy people eat an apple a day.
Keisha eats an apple a day.
 $\therefore$ Keisha is a healthy person.

8. All freshmen must take writing.

Caroline is a freshman.

$\therefore$ Caroline must take writing.

9. All healthy people eat an apple a day.

Herbert is not a healthy person.

$\therefore$ Herbert does not eat an apple a day.

10. If a product of two numbers is 0, then at least one of the numbers is 0.

For a particular number x , neither $(2x + 1)$ nor $(x - 7)$ equals 0.

$\therefore$ The product $(2x + 1)(x - 7)$ is not 0.

Indicate whether the arguments in 21–27 are valid or invalid.
Support your answers by drawing diagrams.

- 21.** All people are mice.
All mice are mortal.
 $\therefore$ All people are mortal.
22. All discrete mathematics students can tell a valid argument from an invalid one.
All thoughtful people can tell a valid argument from an invalid one.
 $\therefore$ All discrete mathematics students are thoughtful.
- 23.** All teachers occasionally make mistakes.
No gods ever make mistakes.
 $\therefore$ No teachers are gods.
24. No vegetarians eat meat.
All vegans are vegetarian.
 $\therefore$ No vegans eat meat.
- 25.** No college cafeteria food is good.
No good food is wasted.
 $\therefore$ No college cafeteria food is wasted.

In exercises 28–32, reorder the premises in each of the arguments to show that the conclusion follows as a valid consequence from the premises. It may be helpful to rewrite the statements in if-then form and replace some statements by their contrapositives. Exercises 28–30 refer to the kinds of Tarski worlds discussed in Example 3.1.13 and 3.3.1. Exercises 31 and 32 are adapted from *Symbolic Logic* by Lewis Carroll.*

- 28.** 1. Every object that is to the right of all the blue objects is above all the triangles.
2. If an object is a circle, then it is to the right of all the blue objects.
3. If an object is not a circle, then it is not gray.
∴ All the gray objects are above all the triangles.

29. 1. All the objects that are to the right of all the triangles are above all the circles.
2. If an object is not above all the black objects, then it is not a square.
3. All the objects that are above all the black objects are to the right of all the triangles.
∴ All the squares are above all the circles.
30. 1. If an object is above all the triangles, then it is above all the blue objects.
2. If an object is not above all the gray objects, then it is not a square.
3. Every black object is a square.
4. Every object that is above all the gray objects is above all the triangles.
∴ If an object is black, then it is above all the blue objects.



Exercise Set 4.1

In 1–3, use the definitions of even, odd, prime, and composite to justify each of your answers.

1. Assume that k is a particular integer.
 - a. Is -17 an odd integer?
 - b. Is 0 an even integer?
 - c. Is $2k - 1$ odd?
2. Assume that m and n are particular integers.
 - a. Is $6m + 8n$ even?
 - b. Is $10mn + 7$ odd?
 - c. If $m > n > 0$, is $m^2 - n^2$ composite?
3. Assume that r and s are particular integers.
 - a. Is $4rs$ even?
 - b. Is $6r + 4s^2 + 3$ odd?
 - c. If r and s are both positive, is $r^2 + 2rs + s^2$ composite?

Disprove the statements in 11–13 by giving a counterexample.

11. For all real numbers a and b , if $a < b$ then $a^2 < b^2$.

Prove the statements in 17 and 18 by the method of exhaustion.

17. Every positive even integer less than 26 can be expressed as a sum of three or fewer perfect squares. (For instance, $10 = 1^2 + 3^2$ and $16 = 4^2$.)



Exercise Set 4.2

9. Assume that a and b are both integers and that $a \neq 0$ and $b \neq 0$. Explain why $(b - a)/(ab^2)$ must be a rational number.



Exercise Set 4.3

4. Does 3 divide $(3k + 1)(3k + 2)(3k + 3)$?

5. Is $6m(2m + 10)$ divisible by 4?

15. For all integers a , b , and c , if $a \mid b$ and $a \mid c$ then $a \mid (b + c)$.

Exercise Set 4.4

For each of the values of n and d given in 1–6, find integers q and r such that $n = dq + r$ and $0 \leq r < d$.

1. $n = 70, d = 9$

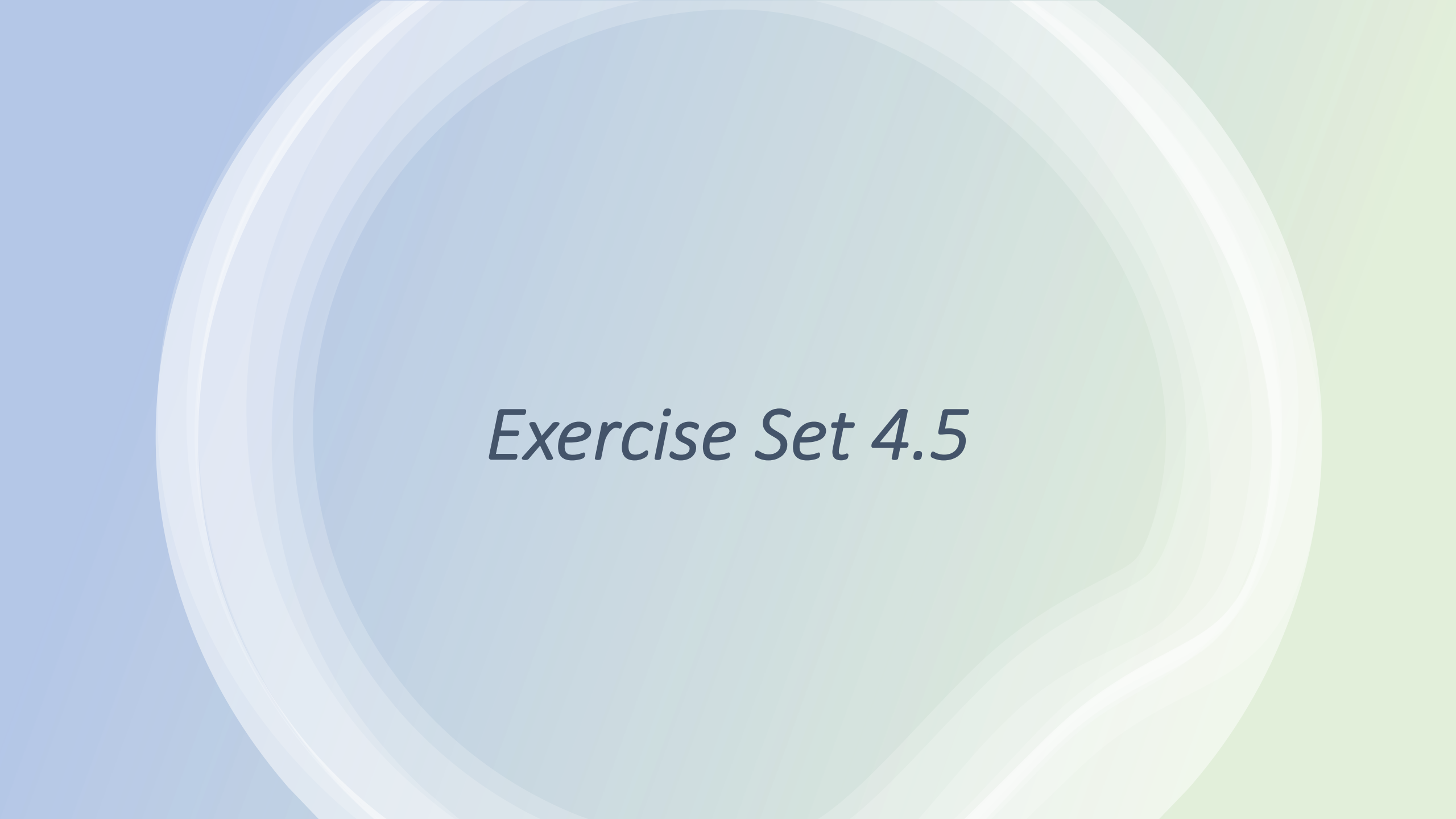
2. $n = 62, d = 7$

3. $n = 36, d = 40$

4. $n = 3, d = 11$

5. $n = -45, d = 11$

6. $n = -27, d = 8$



Exercise Set 4.5

15. For all real numbers x , $\lfloor x - 1 \rfloor = \lfloor x \rfloor - 1$.

Exercise Set 4.6

- 3. Use proof by contradiction to show that for all integers n , $3n + 2$ is not divisible by 3.
- 10. The square root of any irrational number is irrational.

Exercise Set 4.7

16. a. Use proof by contradiction to show that for any integer n , it is impossible for n to equal both $3q_1 + r_1$ and $3q_2 + r_2$, where q_1, q_2, r_1 , and r_2 , are integers, $0 \leq r_1 < 3$, $0 \leq r_2 < 3$, and $r_1 \neq r_2$.
- b. Use proof by contradiction, the quotient-remainder theorem, division into cases, and the result of part (a) to prove that for all integers n , if n^2 is divisible by 3 then n is divisible by 3.
- c. Prove that $\sqrt{3}$ is irrational.



Exercise Set 4.8

Find the greatest common divisor of each of the pairs of integers in 9–12. (Use any method you wish.)

9. 27 and 72

10. 5 and 9

11. 7 and 21

12. 48 and 54

25. Find

a. $\text{lcm}(12, 18)$ **b.** $\text{lcm}(2^2 \cdot 3 \cdot 5, 2^3 \cdot 3^2)$

c. $\text{lcm}(2800, 6125)$

Exercise Set 5.4

1. Suppose $a_1, a_2, a_3, \dots$ is a sequence defined as follows:

$$a_1 = 1, a_2 = 3,$$

$$a_k = a_{k-2} + 2a_{k-1} \quad \text{for all integers } k \geq 3.$$

Prove that a_n is odd for all integers $n \geq 1$.

2. Suppose $b_1, b_2, b_3, \dots$ is a sequence defined as follows:

$$b_1 = 4, b_2 = 12$$

$$b_k = b_{k-2} + b_{k-1} \quad \text{for all integers } k \geq 3.$$

Prove that b_n is divisible by 4 for all integers $n \geq 1$.

3. Suppose that $c_0, c_1, c_2, \dots$ is a sequence defined as follows:

$$c_0 = 2, c_1 = 2, c_2 = 6,$$

$$c_k = 3c_{k-3} \quad \text{for all integers } k \geq 3.$$

Prove that c_n is even for all integers $n \geq 0$.



Exercise Set 8.3



15. Determine which of the following congruence relations are true and which are false.

a. $17 \equiv 2 \pmod{5}$

b. $4 \equiv -5 \pmod{7}$

c. $-2 \equiv -8 \pmod{3}$

d. $-6 \equiv 22 \pmod{2}$



Exercise Set 8.4

3. Let $a = 25$, $b = 19$, and $n = 3$.
- a. Verify that $3 \mid (25 - 19)$.
 - b. Explain why $25 \equiv 19 \pmod{3}$.
 - c. What value of k has the property that $25 = 19 + 3k$?

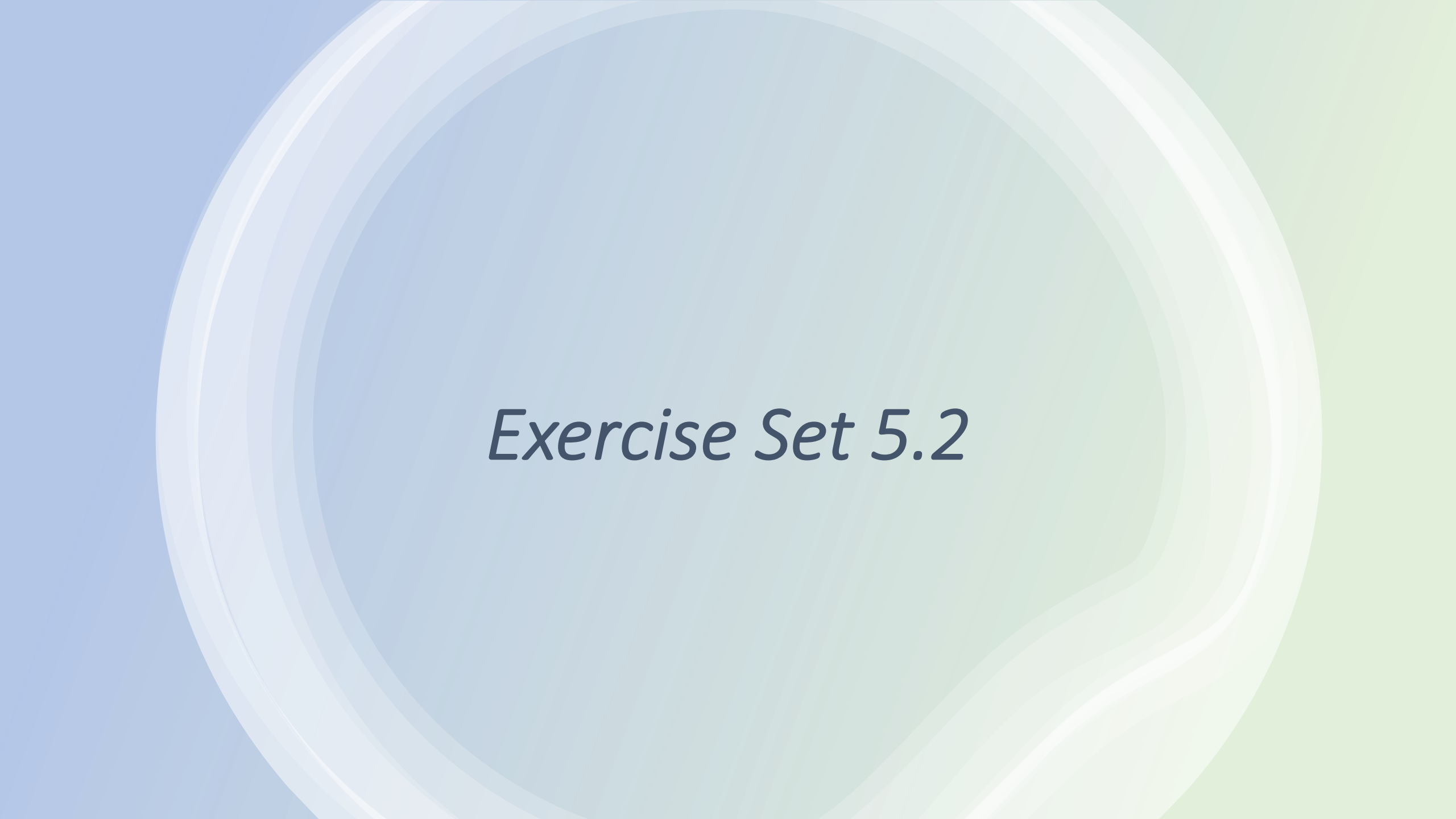
7. Verify the following statements.
- a. $128 \equiv 2 \pmod{7}$ and $61 \equiv 5 \pmod{7}$
 - b. $(128 + 61) \equiv (2 + 5) \pmod{7}$
 - c. $(128 - 61) \equiv (2 - 5) \pmod{7}$
 - d. $(128 \cdot 61) \equiv (2 \cdot 5) \pmod{7}$
 - e. $128^2 \equiv 2^2 \pmod{7}$

In 26 and 27, use the extended Euclidean algorithm to find the greatest common divisor of the given numbers and express it as a linear combination of the two numbers.

26. 6664 and 765

27. 4158 and 1568

- 31.** a. Find an inverse for 210 modulo 13.
b. Find a positive inverse for 210 modulo 13.
c. Find a positive solution for the congruence $210x \equiv 8 \pmod{13}$.



Exercise Set 5.2

3. For each positive integer n , let $P(n)$ be the formula

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

- Write $P(1)$. Is $P(1)$ true?
- Write $P(k)$.
- Write $P(k+1)$.
- In a proof by mathematical induction that the formula holds for all integers $n \geq 1$, what must be shown in the inductive step?



Exercise Set 5.3

3. Observe that

$$\frac{1}{1 \cdot 3} = \frac{1}{3}$$

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} = \frac{2}{5}$$

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} = \frac{3}{7}$$

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \frac{1}{7 \cdot 9} = \frac{4}{9}$$

Guess a general formula and prove it by mathematical induction.

Exercise Set 5.6

Find the first four terms of each of the recursively defined sequences in 1–8.

1. $a_k = 2a_{k-1} + k$, for all integers $k \geq 2$
 $a_1 = 1$

2. $b_k = b_{k-1} + 3k$, for all integers $k \geq 2$
 $b_1 = 1$

3. $c_k = k(c_{k-1})^2$, for all integers $k \geq 1$
 $c_0 = 1$

Exercise Set 5.7

In each of 3–15 a sequence is defined recursively. Use iteration to guess an explicit formula for the sequence. Use the formulas from Section 5.2 to simplify your answers whenever possible.

3. $a_k = ka_{k-1}$, for all integers $k \geq 1$
 $a_0 = 1$

4. $b_k = \frac{b_{k-1}}{1 + b_{k-1}}$, for all integers $k \geq 1$
 $b_0 = 1$

5. $c_k = 3c_{k-1} + 1$, for all integers $k \geq 2$
 $c_1 = 1$

Exercise Set 5.8

In each of 8–10: (a) suppose a sequence of the form $1.t.t^2.t^3 \dots t^n \dots$ where $t \neq 0$, satisfies the given recurrence relation (but not necessarily the initial conditions), and find all possible values of t : (b) suppose a sequence satisfies the given initial conditions as well as the recurrence relation, and find an explicit formula for the sequence.

8. $a_k = 2a_{k-1} + 3a_{k-2}$, for all integers $k \geq 2$
 $a_0 = 1, a_1 = 2$

9. $b_k = 7b_{k-1} - 10b_{k-2}$, for all integers $k \geq 2$
 $b_0 = 2, b_1 = 2$

10. $c_k = c_{k-1} + 6c_{k-2}$, for all integers $k \geq 2$
 $c_0 = 0, c_1 = 3$



Exercise Set 6.1

1. In each of (a)–(f), answer the following questions: Is $A \subseteq B$? Is $B \subseteq A$? Is either A or B a proper subset of the other?

a. $A = \{2, \{2\}, (\sqrt{2})^2\}$, $B = \{2, \{2\}, \{\{2\}\}\}$

b. $A = \{3, \sqrt{5^2 - 4^2}, 24 \bmod 7\}$, $B = \{8 \bmod 5\}$

c. $A = \{\{1, 2\}, \{2, 3\}\}$, $B = \{1, 2, 3\}$

d. $A = \{a, b, c\}$, $B = \{\{a\}, \{b\}, \{c\}\}$

e. $A = \{\sqrt{16}, \{4\}\}$, $B = \{4\}$

f. $A = \{x \in \mathbf{R} \mid \cos x \in \mathbf{Z}\}$, $B = \{x \in \mathbf{R} \mid \sin x \in \mathbf{Z}\}$

2. Complete the proof from Example 6.1.3: Prove that $B \subseteq A$ where

$$A = \{m \in \mathbf{Z} \mid m = 2a \text{ for some integer } a\}$$

and

$$B = \{n \in \mathbf{Z} \mid n = 2b - 2 \text{ for some integer } b\}$$

10. Let $A = \{1, 3, 5, 7, 9\}$, $B = \{3, 6, 9\}$, and $C = \{2, 4, 6, 8\}$.

Find each of the following:

- a. $A \cup B$ b. $A \cap B$ c. $A \cup C$ d. $A \cap C$
e. $A - B$ f. $B - A$ g. $B \cup C$ h. $B \cap C$

11. Let the universal set be the set $\mathbf{R}$ of all real numbers and let

$A = \{x \in \mathbf{R} \mid 0 < x \leq 2\}$, $B = \{x \in \mathbf{R} \mid 1 \leq x < 4\}$, and
 $C = \{x \in \mathbf{R} \mid 3 \leq x < 9\}$. Find each of the following:

- a. $A \cup B$ b. $A \cap B$ c. A^c d. $A \cup C$
e. $A \cap C$ f. B^c g. $A^c \cap B^c$
h. $A^c \cup B^c$ i. $(A \cap B)^c$ j. $(A \cup B)^c$

12. Let the universal set be the set $\mathbf{R}$ of all real numbers and

let $A = \{x \in \mathbf{R} \mid -3 \leq x \leq 0\}$, $B = \{x \in \mathbf{R} \mid -1 < x < 2\}$,
and $C = \{x \in \mathbf{R} \mid 6 < x \leq 8\}$. Find each of the following:

- a. $A \cup B$ b. $A \cap B$ c. A^c d. $A \cup C$
e. $A \cap C$ f. B^c g. $A^c \cap B^c$
h. $A^c \cup B^c$ i. $(A \cap B)^c$ j. $(A \cup B)^c$

15. Draw Venn diagrams to describe sets A , B , and C that satisfy the given conditions.

a. $A \cap B = \emptyset$, $A \subseteq C$, $C \cap B \neq \emptyset$

b. $A \subseteq B$, $C \subseteq B$, $A \cap C \neq \emptyset$

c. $A \cap B \neq \emptyset$, $B \cap C \neq \emptyset$, $A \cap C = \emptyset$, $A \not\subseteq B$, $C \not\subseteq B$

16. Let $A = \{a, b, c\}$, $B = \{b, c, d\}$, and $C = \{b, c, e\}$.

a. Find $A \cup (B \cap C)$, $(A \cup B) \cap C$, and $(A \cup B) \cap (A \cup C)$. Which of these sets are equal?

b. Find $A \cap (B \cup C)$, $(A \cap B) \cup C$, and $(A \cap B) \cup (A \cap C)$. Which of these sets are equal?

c. Find $(A - B) - C$ and $A - (B - C)$. Are these sets equal?

27. **a.** Is $\{\{a, d, e\}, \{b, c\}, \{d, f\}\}$ a partition of $\{a, b, c, d, e, f\}$?
- b. Is $\{\{w, x, v\}, \{u, y, q\}, \{p, z\}\}$ a partition of $\{p, q, u, v, w, x, y, z\}$?
- c. Is $\{\{5, 4\}, \{7, 2\}, \{1, 3, 4\}, \{6, 8\}\}$ a partition of $\{1, 2, 3, 4, 5, 6, 7, 8\}$?
- d.** Is $\{\{3, 7, 8\}, \{2, 9\}, \{1, 4, 5\}\}$ a partition of $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$?
- e. Is $\{\{1, 5\}, \{4, 7\}, \{2, 8, 6, 3\}\}$ a partition of $\{1, 2, 3, 4, 5, 6, 7, 8\}$?

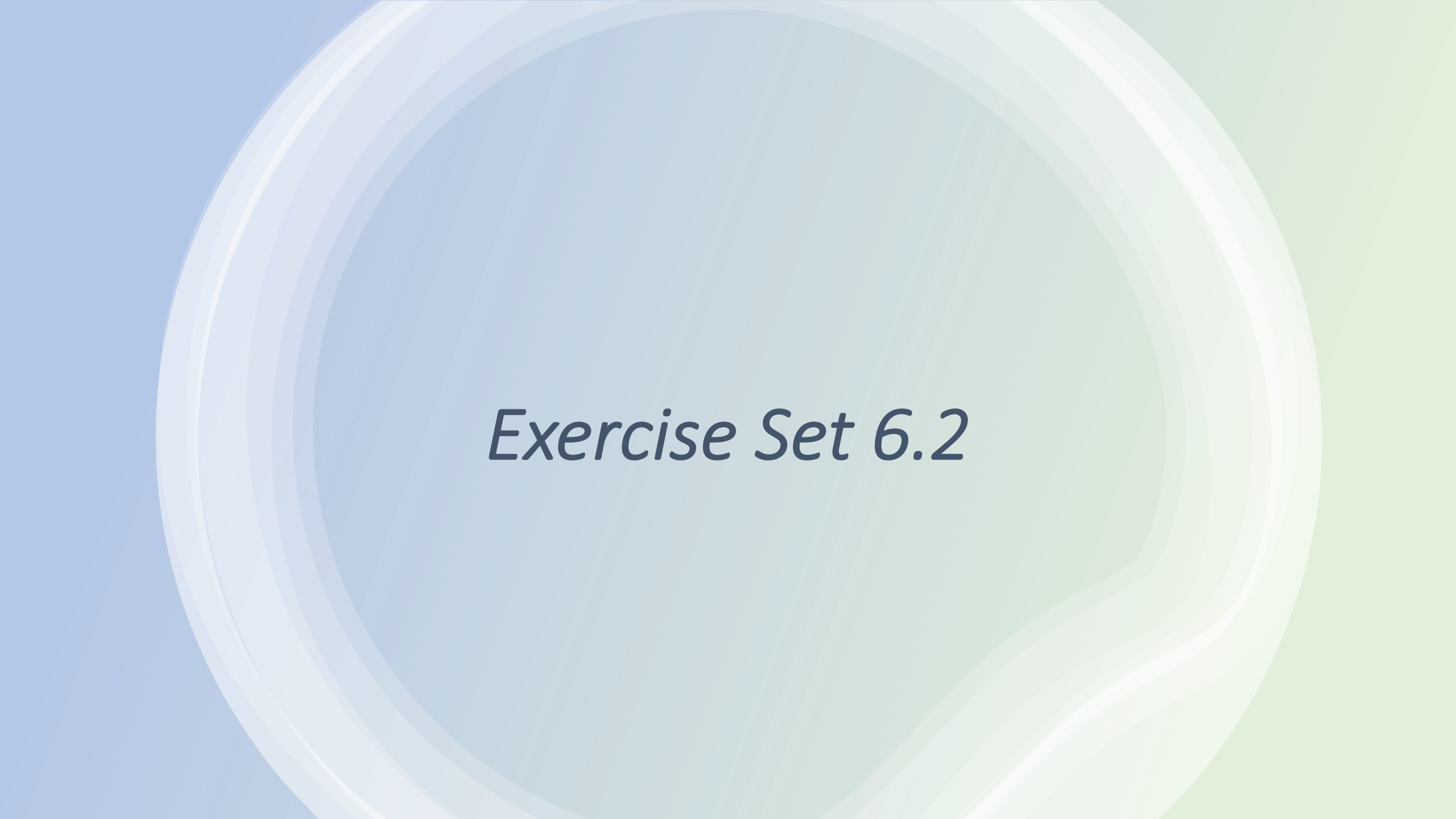
31. Suppose $A = \{1, 2\}$ and $B = \{2, 3\}$. Find each of the following:

a. $\mathcal{P}(A \cap B)$

b. $\mathcal{P}(A)$

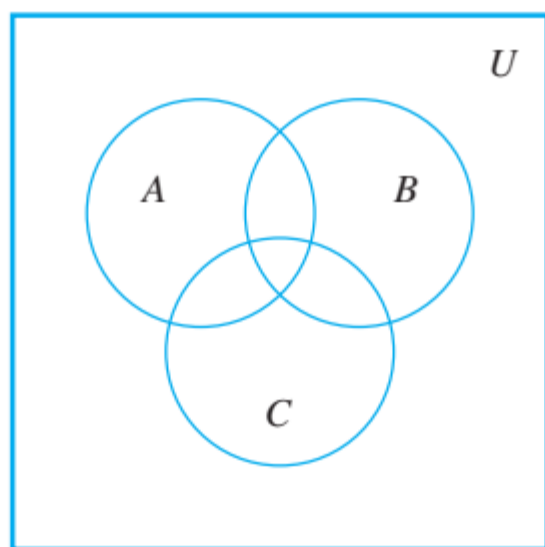
c. $\mathcal{P}(A \cup B)$

d. $\mathcal{P}(A \times B)$



Exercise Set 6.2

23. Consider the Venn diagram below.



- a. Illustrate one of the distributive laws by shading in the region corresponding to $A \cup (B \cap C)$ on one copy of the diagram and $(A \cup B) \cap (A \cup C)$ on another.
- b. Illustrate the other distributive law by shading in the region corresponding to $A \cap (B \cup C)$ on one copy of the diagram and $(A \cap B) \cup (A \cap C)$ on another.
- c. Illustrate one of De Morgan's laws by shading in the region corresponding to $(A \cup B)^c$ on one copy of the diagram and $A^c \cap B^c$ on the other. (Leave the set C out of your diagrams.)
- d. Illustrate the other De Morgan's law by shading in the region corresponding to $(A \cap B)^c$ on one copy of the diagram and $A^c \cup B^c$ on the other. (Leave the set C out of your diagrams.)



Exercise Set 6.3

For each of 1–4 find a counterexample to show that the statement is false. Assume all sets are subsets of a universal set U .

1. For all sets A , B , and C , $(A \cap B) \cup C = A \cap (B \cup C)$.
2. For all sets A and B , $(A \cup B)^c = A^c \cup B^c$.
3. For all sets A , B , and C , if $A \not\subseteq B$ and $B \not\subseteq C$ then $A \not\subseteq C$.
4. For all sets A , B , and C , if $B \cap C \subseteq A$ then
$$(A - B) \cap (A - C) = \emptyset.$$



Exercise Set 1.2

8. Let $A = \{c, d, f, g\}$, $B = \{f, j\}$, and $C = \{d, g\}$. Answer each of the following questions. Give reasons for your answers.

- a. Is $B \subseteq A$? b. Is $C \subseteq A$?
b. Is $C \subseteq C$? d. Is C a proper subset of A ?

9. a. Is $3 \in \{1, 2, 3\}$? b. Is $1 \subseteq \{1\}$?
c. Is $\{2\} \in \{1, 2\}$? d. Is $\{3\} \in \{1, \{2\}, \{3\}\}$?
e. Is $1 \in \{1\}$? f. Is $\{2\} \subseteq \{1, \{2\}, \{3\}\}$?
g. Is $\{1\} \subseteq \{1, 2\}$? h. Is $1 \in \{\{1\}, 2\}$?
i. Is $\{1\} \subseteq \{1, \{2\}\}$? j. Is $\{1\} \subseteq \{1\}$?



Exercise Set 1.3

1. Let $A = \{2, 3, 4\}$ and $B = \{6, 8, 10\}$ and define a relation R from A to B as follows: For all $(x, y) \in A \times B$,

$$(x, y) \in R \quad \text{means that} \quad \frac{y}{x} \text{ is an integer.}$$

- Is $4 R 6$? Is $4 R 8$? Is $(3, 8) \in R$? Is $(2, 10) \in R$?
- Write R as a set of ordered pairs.
- Write the domain and co-domain of R .
- Draw an arrow diagram for R .

3. Let $E = \{1, 2, 3\}$ and $F = \{-2, -1, 0\}$ and define a relation T from E to F as follows: For all $(x, y) \in E \times F$,

$$(x, y) \in T \quad \text{means that} \quad \frac{x - y}{3} \text{ is an integer.}$$

- Is $3 T 0$? Is $1 T (-1)$? Is $(2, -1) \in T$? Is $(3, -2) \in T$?
- Write T as a set of ordered pairs.
- Write the domain and co-domain of T .
- Draw an arrow diagram for T .

5. Define a relation S from $\mathbf{R}$ to $\mathbf{R}$ as follows:

For all $(x, y) \in \mathbf{R} \times \mathbf{R}$,

$$(x, y) \in S \quad \text{means that} \quad x \geq y.$$

- a. Is $(2, 1) \in S$? Is $(2, 2) \in S$? Is $2 S 3$? Is $(-1) S (-2)$?
- b. Draw the graph of S in the Cartesian plane.



Exercise Set 8.1

3. The **congruence modulo 3** relation, T , is defined from $\mathbf{Z}$ to $\mathbf{Z}$ as follows: For all integers m and n ,

$$m T n \iff 3 \mid (m - n).$$

- a. Is $10 T 1$? Is $1 T 10$? Is $(2, 2) \in T$? Is $(8, 1) \in T$?
- b. List five integers n such that $n T 0$.
- c. List five integers n such that $n T 1$.
- d. List five integers n such that $n T 2$.
- H** e. Make and prove a conjecture about which integers are related by T to 0, which integers are related by T to 1, and which integers are related by T to 2.

5. Let $X = \{a, b, c\}$. Recall that $\mathcal{P}(X)$ is the power set of X . Define a relation $\mathbf{R}$ on $\mathcal{P}(X)$ as follows:
For all $A, B \in \mathcal{P}(X)$,

$$A \mathbf{R} B \iff A \text{ has the same number of elements as } B.$$

- a. Is $\{a, b\} \mathbf{R} \{b, c\}$? b. Is $\{a\} \mathbf{R} \{a, b\}$?
c. Is $\{c\} \mathbf{R} \{b\}$?
7. Define a relation R on $\mathbf{Z}$ as follows: For all integers m and n ,

$$m R n \iff 5 \mid (m^2 - n^2).$$

- a. Is $1 R (-9)$? b. Is $2 R 13$?
c. Is $2 R (-8)$? d. Is $(-8) R 2$?

- 10.** Let $A = \{3, 4, 5\}$ and $B = \{4, 5, 6\}$ and let R be the “less than” relation. That is, for all $(x, y) \in A \times B$,

$$x R y \iff x < y.$$

State explicitly which ordered pairs are in R and R^{-1} .

Draw the directed graphs of the relations defined in 13–18.

13. Define a relation R on $A = \{0, 1, 2, 3\}$ by $R = \{(0, 0), (1, 2), (2, 2)\}$.

15. Let $A = \{2, 3, 4, 5, 6, 7, 8\}$ and define a relation R on A as follows: For all $x, y \in A$,

$$x R y \iff x \mid y.$$

16. Let $A = \{5, 6, 7, 8, 9, 10\}$ and define a relation S on A as follows: For all $x, y \in A$,

$$x S y \iff 2 \mid (x - y).$$

Exercises 19–20 refer to unions and intersections of relations. Since relations are subsets of Cartesian products, their unions and intersections can be calculated as for any subsets. Given two relations R and S from A to B ,

$$R \cup S = \{(x, y) \in A \times B \mid (x, y) \in R \text{ or } (x, y) \in S\}$$

$$R \cap S = \{(x, y) \in A \times B \mid (x, y) \in R \text{ and } (x, y) \in S\}.$$

- 19.** Let $A = \{2, 4\}$ and $B = \{6, 8, 10\}$ and define relations R and S from A to B as follows: For all $(x, y) \in A \times B$,

$$x R y \iff x \mid y \text{ and}$$

$$x S y \iff y - 4 = x.$$

State explicitly which ordered pairs are in $A \times B$, R , S , $R \cup S$, and $R \cap S$.



Exercise Set 8.2

In 1–8 a number of relations are defined on the set $A = \{0, 1, 2, 3\}$. For each relation:

- Draw the directed graph.
- Determine whether the relation is reflexive.
- Determine whether the relation is symmetric.
- Determine whether the relation is transitive.

Give a counterexample in each case in which the relation does not satisfy one of the properties.

1. $R_1 = \{(0, 0), (0, 1), (0, 3), (1, 1), (1, 0), (2, 3), (3, 3)\}$

2. $R_2 = \{(0, 0), (0, 1), (1, 1), (1, 2), (2, 2), (2, 3)\}$

3. $R_3 = \{(2, 3), (3, 2)\}$

In 9–33 determine whether the given relation is reflexive, symmetric, transitive, or none of these. Justify your answers.

9. R is the “greater than or equal to” relation on the set of real numbers: For all $x, y \in \mathbf{R}$, $x R y \Leftrightarrow x \geq y$.
10. C is the circle relation on the set of real numbers: For all $x, y \in \mathbf{R}$, $x C y \Leftrightarrow x^2 + y^2 = 1$.
11. D is the relation defined on $\mathbf{R}$ as follows: For all $x, y \in \mathbf{R}$, $x D y \Leftrightarrow xy \geq 0$.
12. E is the congruence modulo 2 relation on $\mathbf{Z}$: For all $m, n \in \mathbf{Z}$, $m E n \Leftrightarrow 2 \mid (m - n)$.

In 51–53. R , S , and T are relations defined on $A = \{0, 1, 2, 3\}$.

- 51.** Let $R = \{(0, 1), (0, 2), (1, 1), (1, 3), (2, 2), (3, 0)\}$. Find R' , the transitive closure of R .
52. Let $S = \{(0, 0), (0, 3), (1, 0), (1, 2), (2, 0), (3, 2)\}$. Find S' , the transitive closure of S .



Exercise Set 8.3

In each of 3–14, the relation R is an equivalence relation on the set A . Find the distinct equivalence classes of R .

3. $A = \{0, 1, 2, 3, 4\}$
 $R = \{(0, 0), (0, 4), (1, 1), (1, 3), (2, 2), (3, 1), (3, 3),$
 $(4, 0), (4, 4)\}$

4. $A = \{a, b, c, d\}$
 $R = \{(a, a), (b, b), (b, d), (c, c), (d, b), (d, d)\}$

5. $A = \{1, 2, 3, 4, \dots, 20\}$. R is defined on A as follows:

$$\text{For all } x, y \in A, \quad x R y \iff 4 \mid (x - y).$$

In 19–31, (1) prove that the relation is an equivalence relation, and (2) describe the distinct equivalence classes of each relation.

19. A is the set of all students at your college.

a. R is the relation defined on A as follows: For all x and y in A ,

$$x R y \iff x \text{ has the same major (or double major) as } y.$$

(Assume “undeclared” is a major.)

b. S is the relation defined on A as follows: For all $x, y \in A$,

$$x S y \iff x \text{ is the same age as } y.$$

20. E is the relation defined on $\mathbf{Z}$ as follows:

$$\text{For all } m, n \in \mathbf{Z}, \quad m E n \iff 2 \mid (m - n).$$

Exercise Set 8.5

1. Each of the following is a relation on $\{0, 1, 2, 3\}$. Draw directed graphs for each relation, and indicate which relations are antisymmetric.

a. $R_1 = \{(0, 0), (0, 2), (1, 0), (1, 3), (2, 2), (3, 0), (3, 1)\}$

b. $R_2 = \{(0, 1), (0, 2), (1, 1), (1, 2), (1, 3), (2, 2), (3, 2)\}$

c. $R_3 = \{(0, 0), (0, 3), (1, 0), (1, 3), (2, 2), (3, 3), (3, 2)\}$

d. $R_4 = \{(0, 0), (1, 0), (1, 2), (1, 3), (2, 0), (2, 1), (3, 2), (3, 0)\}$

2. Let P be the set of all people in the world and define a relation R on P as follows: For all $x, y \in P$,

$$x R y \iff x \text{ is no older than } y.$$

5. Let $\mathbf{R}$ be the set of all real numbers and define a relation R on $\mathbf{R} \times \mathbf{R}$ as follows: For all (a, b) and (c, d) in $\mathbf{R} \times \mathbf{R}$,

$$(a, b) R (c, d) \iff \begin{array}{l} \text{either } a < c \text{ or both } a = c \\ \text{and } b \leq d. \end{array}$$

Is R a partial order relation? Prove or give a counter-example.

16. Consider the “divides” relation on each of the following sets A . Draw the Hasse diagram for each relation.
- a. $A = \{1, 2, 4, 5, 10, 15, 20\}$
 - b. $A = \{2, 3, 4, 6, 8, 9, 12, 18\}$
17. Consider the “subset” relation on $\mathcal{P}(S)$ for each of the following sets S . Draw the Hasse diagram for each relation.
- a. $S = \{0, 1\}$
 - b. $S = \{0, 1, 2\}$

In 22–29, find all greatest, least, maximal, and minimal elements for the relations in each of the referenced exercises.

22. Exercise 16(a)

23. Exercise 16(b)

24. Exercise 17(a)

25. Exercise 17(b)

31. Let $A = \{a, b, c, d\}$, and let R be the relation

$$R = \{(a, a), (b, b), (c, c), (d, d), (c, a), (a, d), \\ (c, d), (b, c), (b, d), (b, a)\}.$$

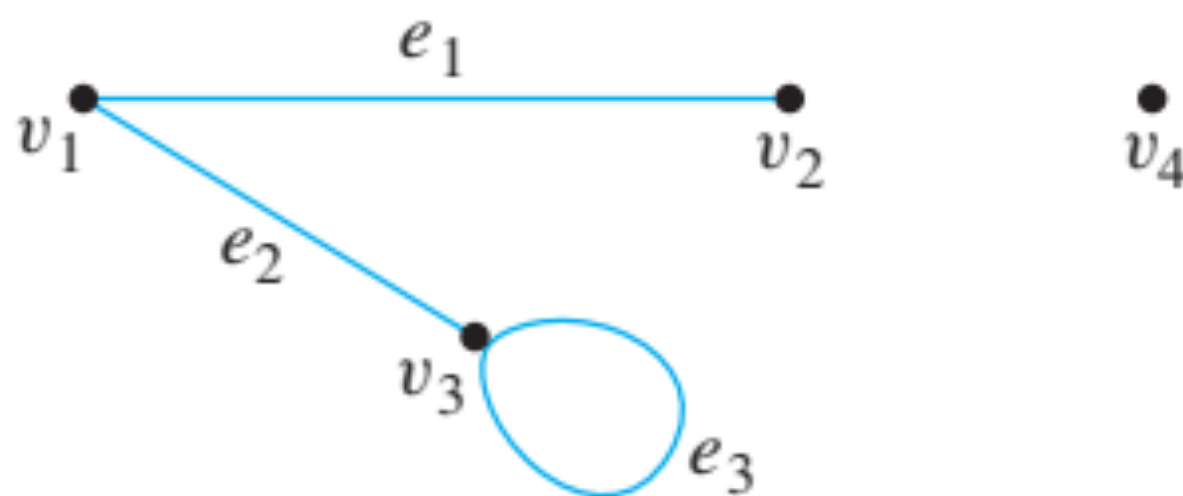
Is R a total order on A ? Justify your answer.



Exercise Set 10.1

In 1 and 2, graphs are represented by drawings. Define each graph formally by specifying its vertex set, its edge set, and a table giving the edge-endpoint function.

1.

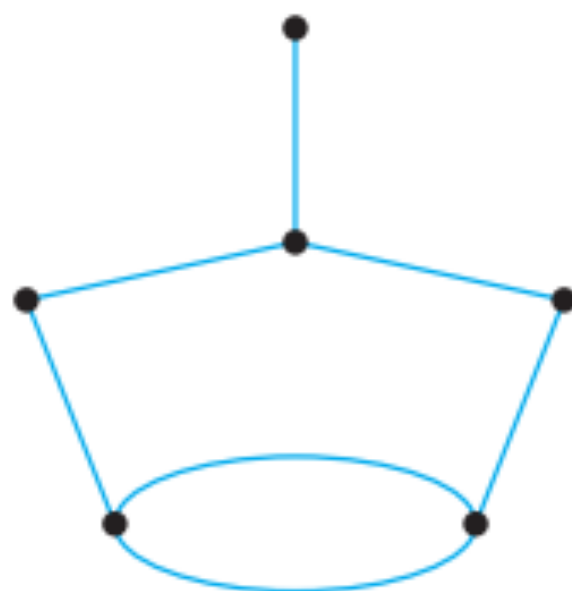
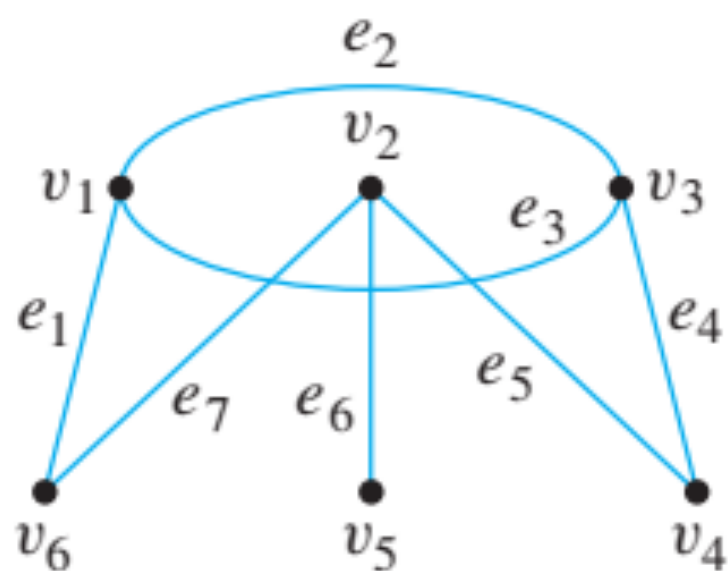


3. Graph G has vertex set $\{v_1, v_2, v_3, v_4, v_5\}$ and edge set $\{e_1, e_2, e_3, e_4\}$, with edge-endpoint function as follows:

Edge	Endpoints
e_1	$\{v_1, v_2\}$
e_2	$\{v_1, v_2\}$
e_3	$\{v_2, v_3\}$
e_4	$\{v_2\}$

In 5–7, show that the two drawings represent the same graph by labeling the vertices and edges of the right-hand drawing to correspond to those of the left-hand drawing.

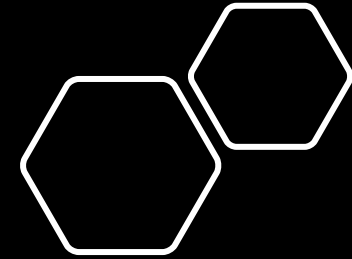
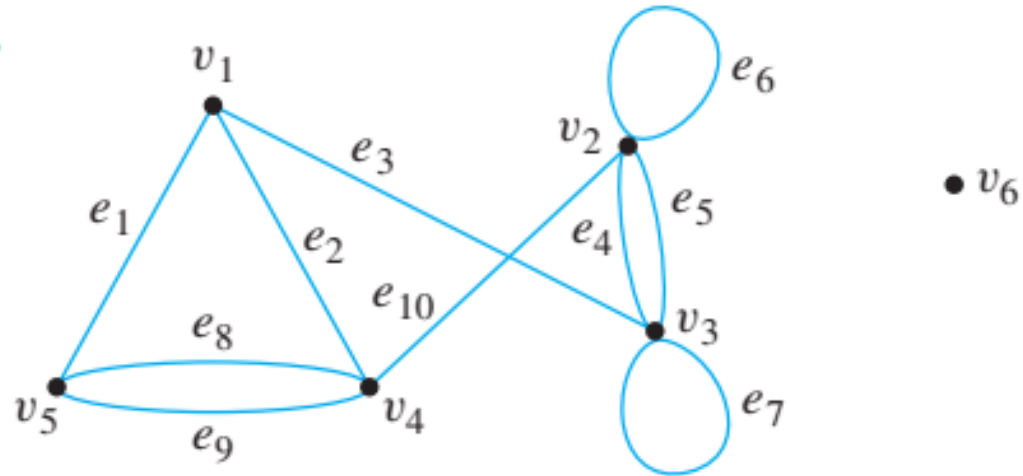
5.



For each of the graphs in 8 and 9:

- (i) Find all edges that are incident on v_1 .
- (ii) Find all vertices that are adjacent to v_3 .
- (iii) Find all edges that are adjacent to e_1 .
- (iv) Find all loops.
- (v) Find all parallel edges.
- (vi) Find all isolated vertices.
- (vii) Find the degree of v_3 .
- (viii) Find the total degree of the graph.

8.



In each of 17–25, either draw a graph with the specified properties or explain why no such graph exists.

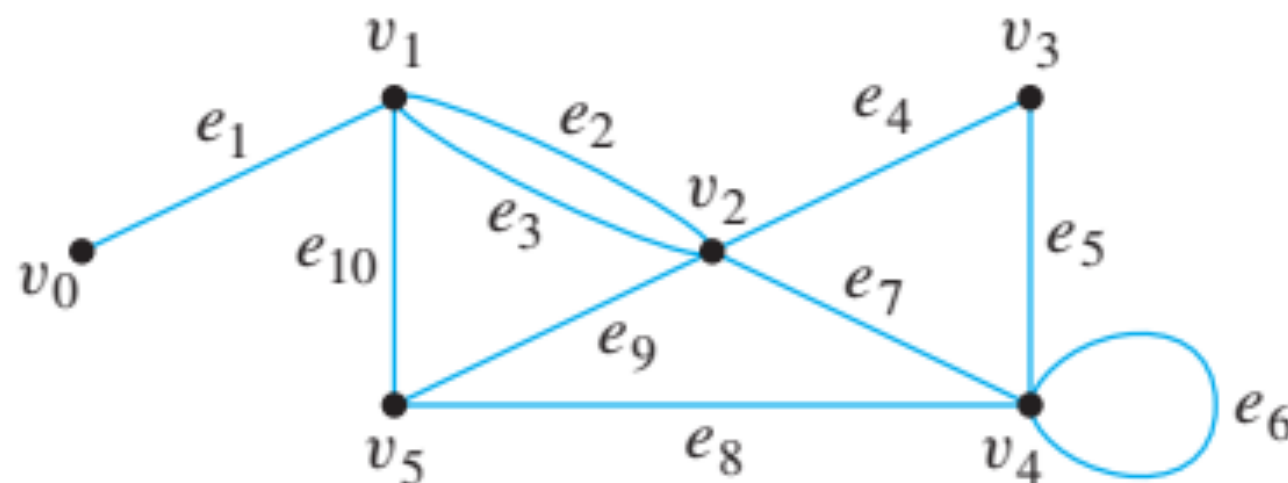
17. Graph with five vertices of degrees 1, 2, 3, 3, and 5.

18. Graph with four vertices of degrees 1, 2, 3, and 3.

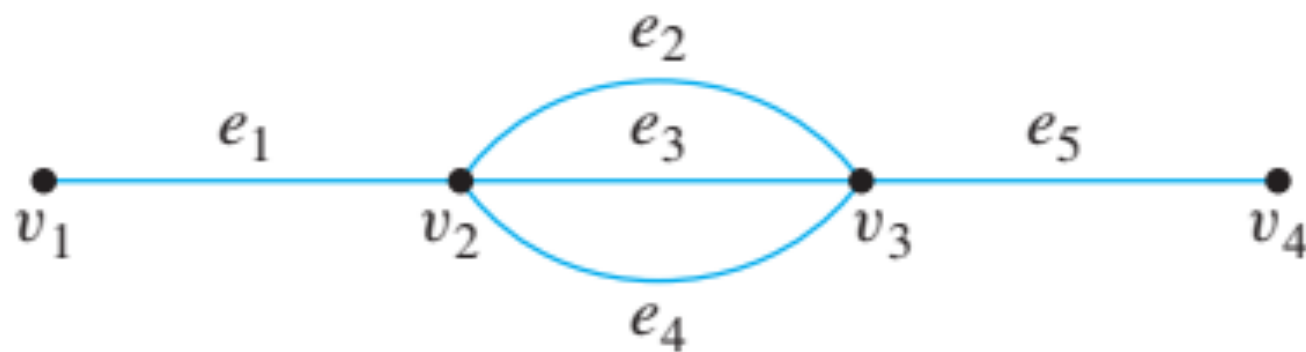
Exercise Set 10.2

1. In the graph below, determine whether the following walks are trails, paths, closed walks, circuits, simple circuits, or just walks.

- a. $v_0e_1v_1e_{10}v_5e_9v_2e_2v_1$ b. $v_4e_7v_2e_9v_5e_{10}v_1e_3v_2e_9v_5$
c. v_2 d. $v_5v_2v_3v_4v_4v_5$
e. $v_2v_3v_4v_5v_2v_4v_3v_2$ f. $e_5e_8e_{10}e_3$



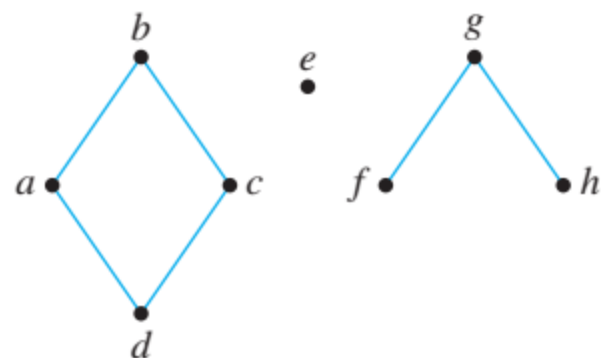
4. Consider the following graph.



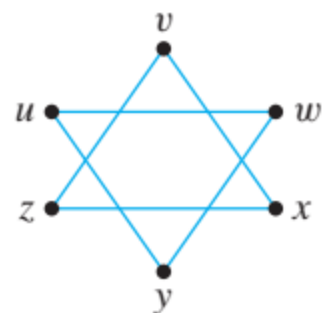
- a. How many paths are there from v_1 to v_4 ?
- b. How many trails are there from v_1 to v_4 ?
- c. How many walks are there from v_1 to v_4 ?

8. Find the number of connected components for each of the following graphs.

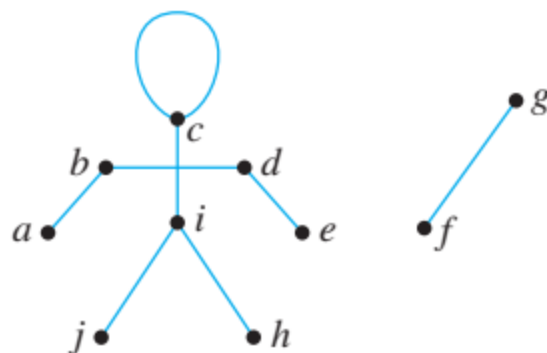
a.



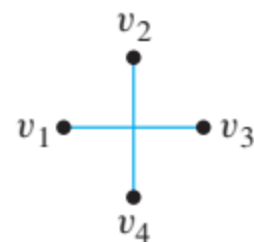
b.



c.

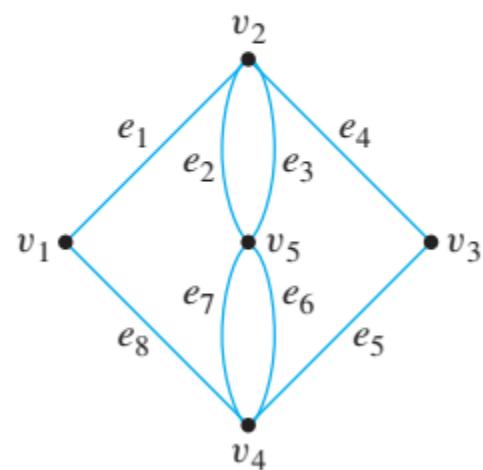


d.

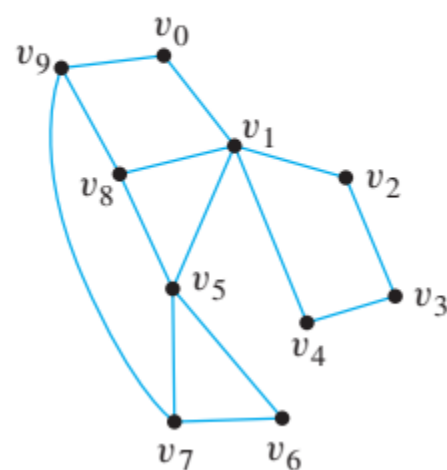


Determine which of the graphs in 12–17 have Euler circuits. If the graph does not have an Euler circuit, explain why not. If it does have an Euler circuit, describe one.

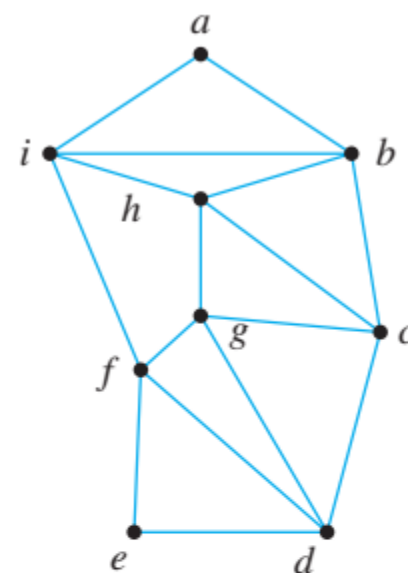
12.



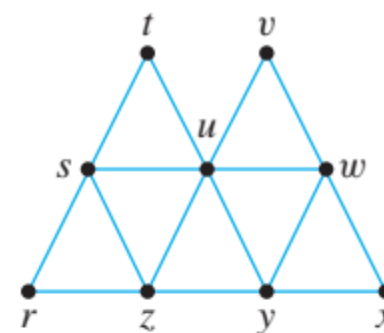
13.



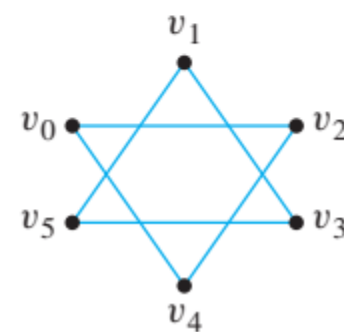
14.



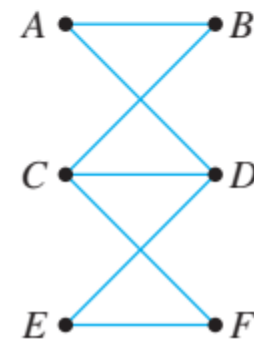
15.



16.

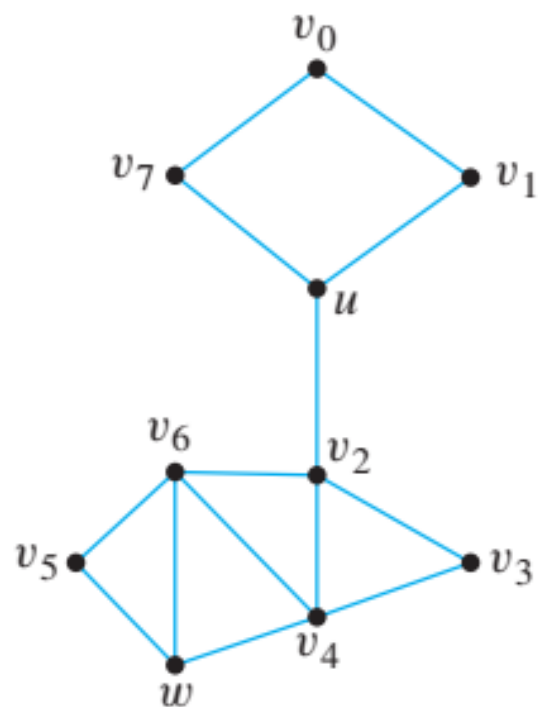


17.

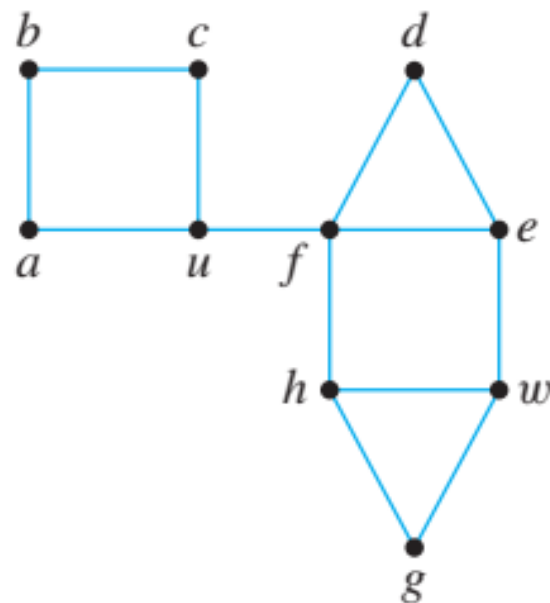


For each of the graphs in 19–21, determine whether there is an Euler path from u to w . If there is, find such a path.

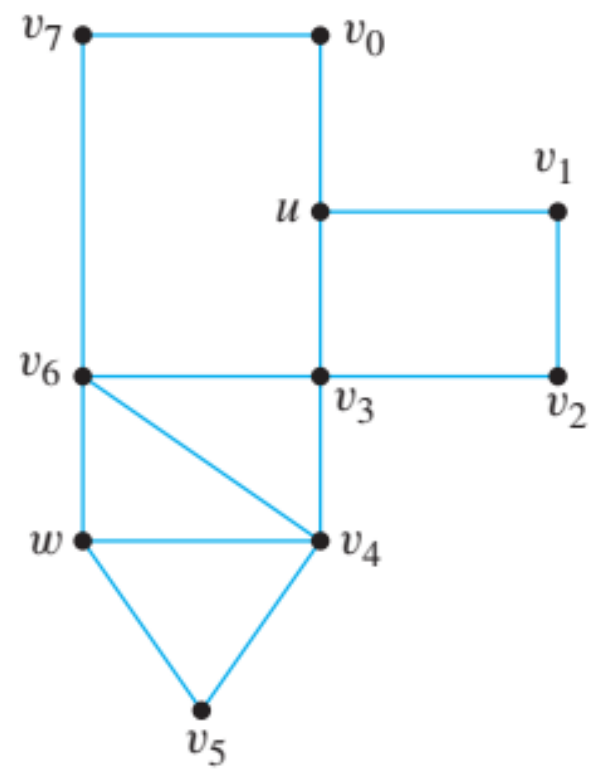
19.



20.

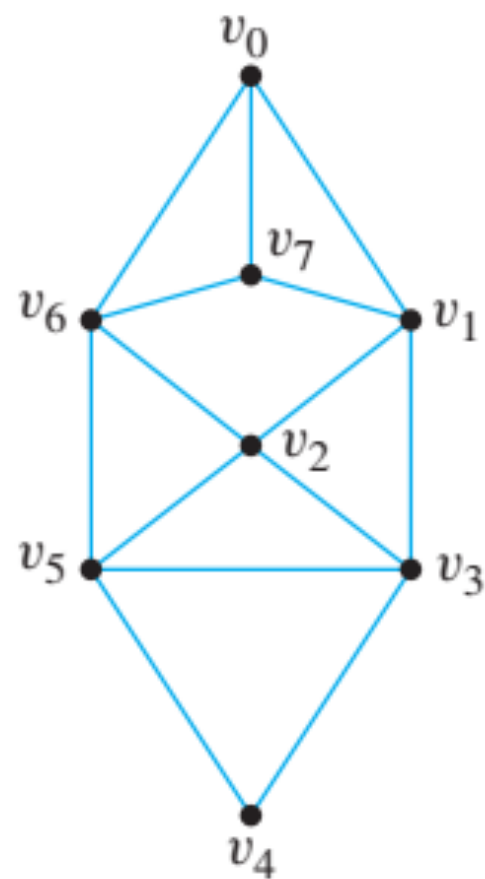


21.

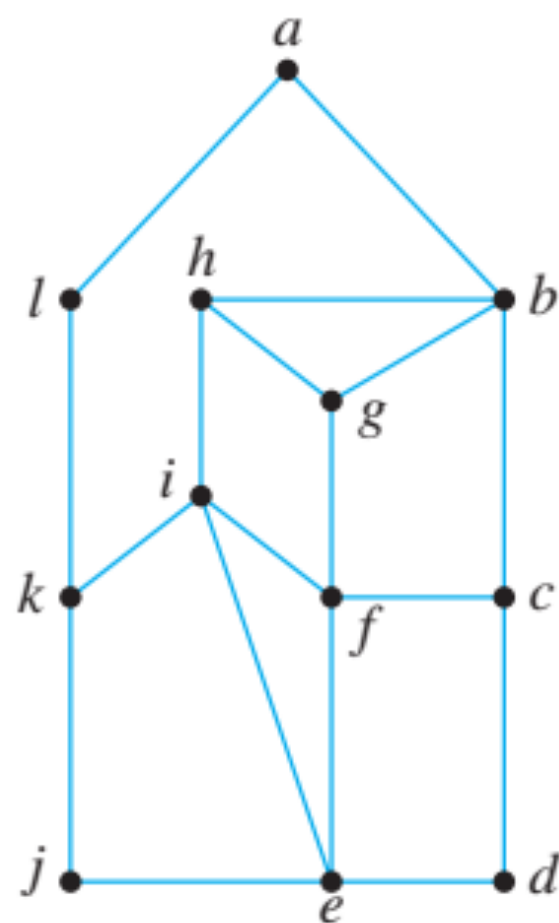


Find Hamiltonian circuits for each of the graphs in 23 and 24.

23.



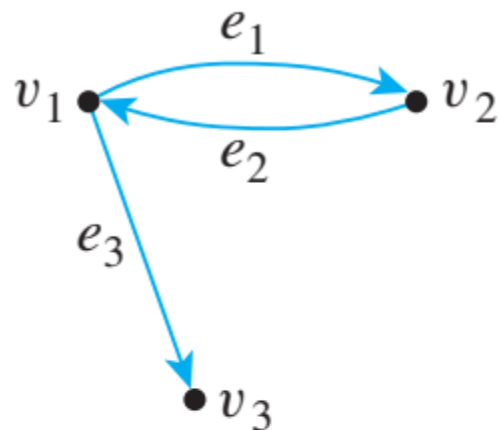
24.



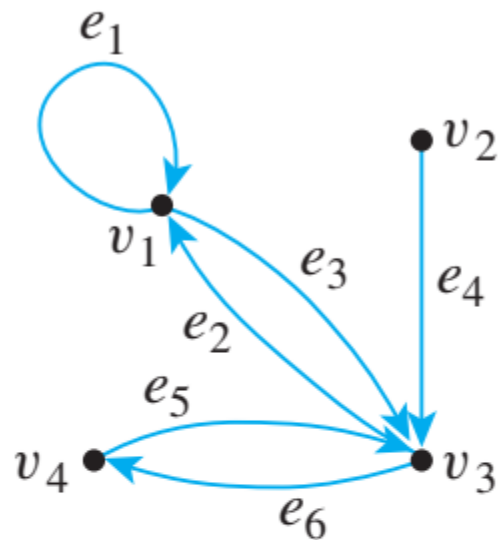
Exercise Set 10.3

2. Find the adjacency matrices for the following directed graphs.

a.



b.

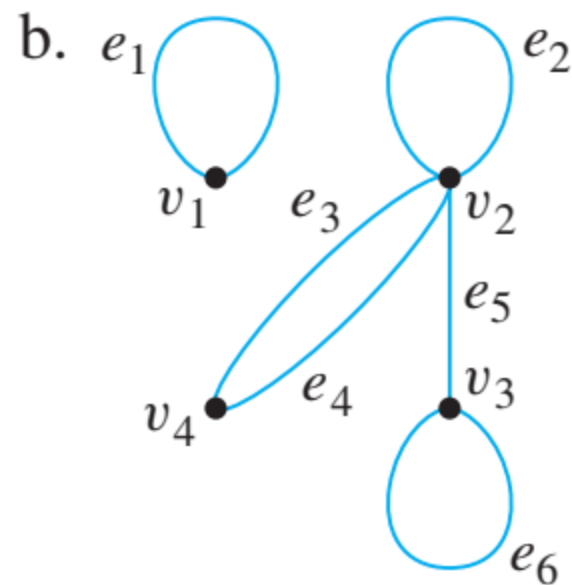
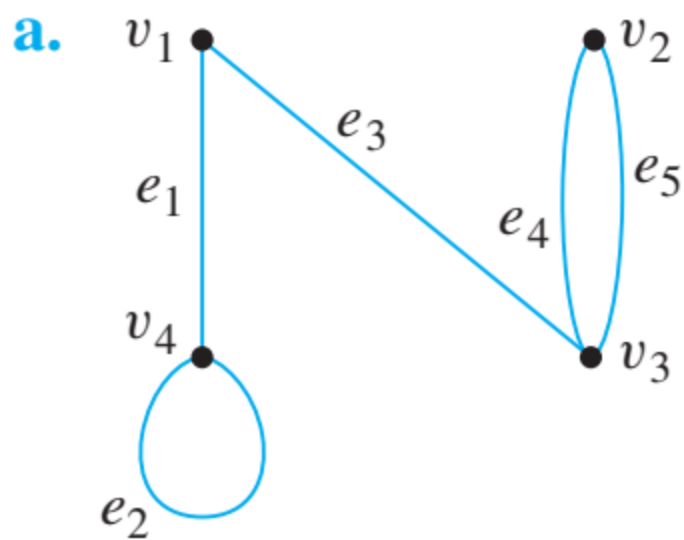


3. Find directed graphs that have the following adjacency matrices:

a.
$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

b.
$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

4. Find adjacency matrices for the following (undirected) graphs.



c. K_4 , the complete graph on four vertices

d. $K_{2,3}$, the complete bipartite graph on (2, 3) vertices

5. Find graphs that have the following adjacency matrices.

a.
$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix}$$

b.
$$\begin{bmatrix} 0 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

19. **a.** Let $\mathbf{A} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$. Find $\mathbf{A}^2$ and $\mathbf{A}^3$.

- b. Let G be the graph with vertices v_1 , v_2 , and v_3 and with $\mathbf{A}$ as its adjacency matrix. Use the answers to part (a) to find the number of walks of length 2 from v_1 to v_3 and the number of walks of length 3 from v_1 to v_3 . Do not draw G to solve this problem.
- c. Examine the calculations you performed in answering part (a) to find five walks of length 2 from v_3 to v_3 . Then draw G and find the walks by visual inspection.

20. The following is an adjacency matrix for a graph:

$$\begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} v_1 \quad v_2 \quad v_3 \quad v_4 \\ \left[\begin{array}{cccc} 0 & 1 & 1 & 0 \\ 1 & 0 & 2 & 1 \\ 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{array} \right] \end{array}$$

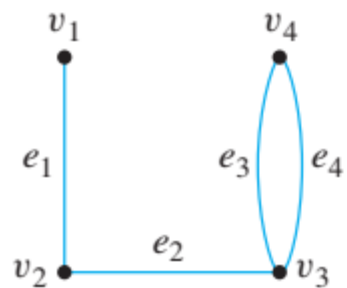
Answer the following questions by examining the matrix and its powers only, not by drawing the graph:

- How many walks of length 2 are there from v_2 to v_3 ?
- How many walks of length 2 are there from v_3 to v_4 ?
- How many walks of length 3 are there from v_1 to v_4 ?
- How many walks of length 3 are there from v_2 to v_3 ?

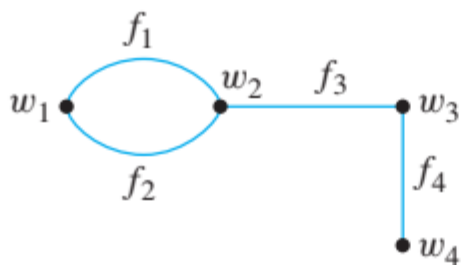
Exercise Set 10.4

For each pair of graphs G and G' in 1–5, determine whether G and G' are isomorphic. If they are, give functions $g: V(G) \rightarrow V(G')$ and $h: E(G) \rightarrow E(G')$ that define the isomorphism. If they are not, give an invariant for graph isomorphism that they do not share.

1.

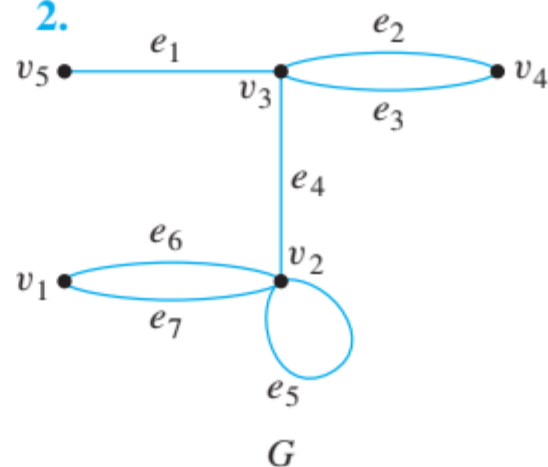


G

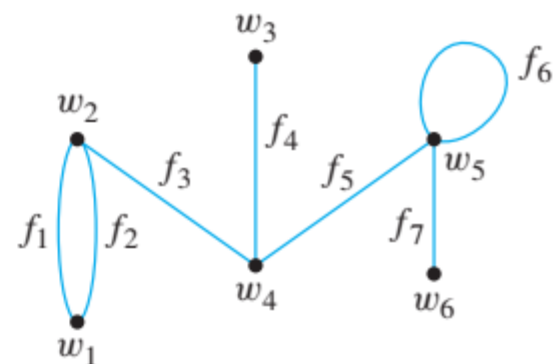


G'

2.



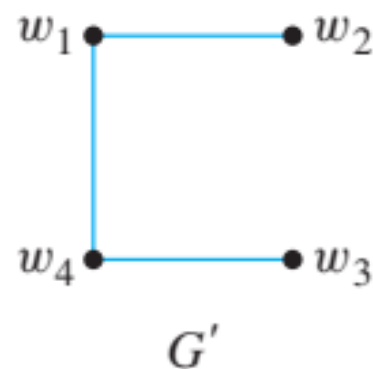
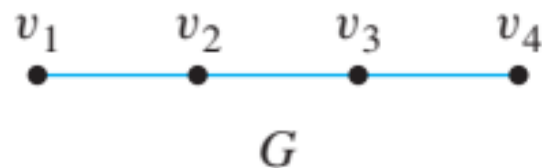
G



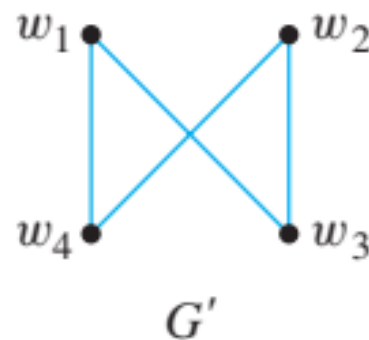
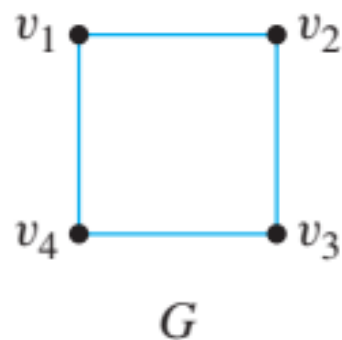
G'

For each pair of simple graphs G and G' in 6–13, determine whether G and G' are isomorphic. If they are, give a function $g: V(G) \rightarrow V(G')$ that defines the isomorphism. If they are not, give an invariant for graph isomorphism that they do not share.

6.



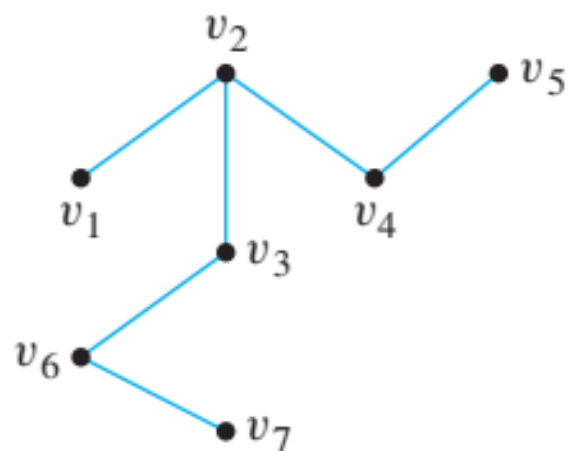
7.



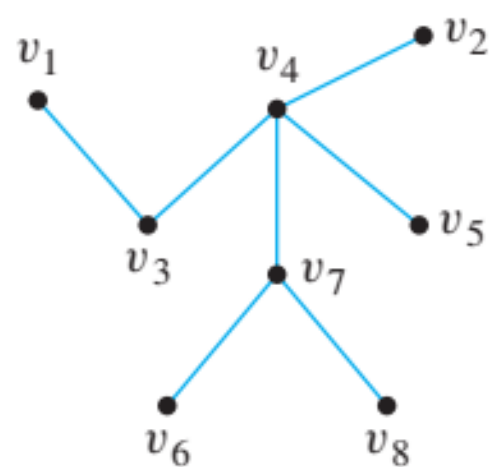
Exercise Set 10.5

7. Find all terminal vertices and all internal vertices for the following trees.

a.



b.

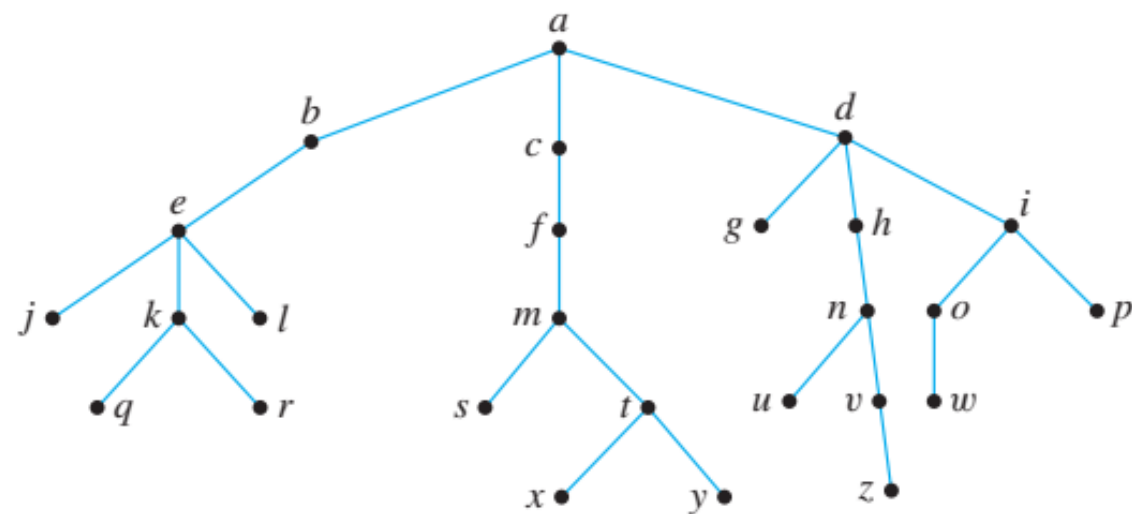


In each of 8–21, either draw a graph with the given specifications or explain why no such graph exists.

- 8. Tree, nine vertices, nine edges
- 9. Graph, connected, nine vertices, nine edges
- 10. Graph, circuit-free, nine vertices, six edges
- 11. Tree, six vertices, total degree 14
- 12. Tree, five vertices, total degree 8
- 13. Graph, connected, six vertices, five edges, has a nontrivial circuit

Exercise Set 10.6

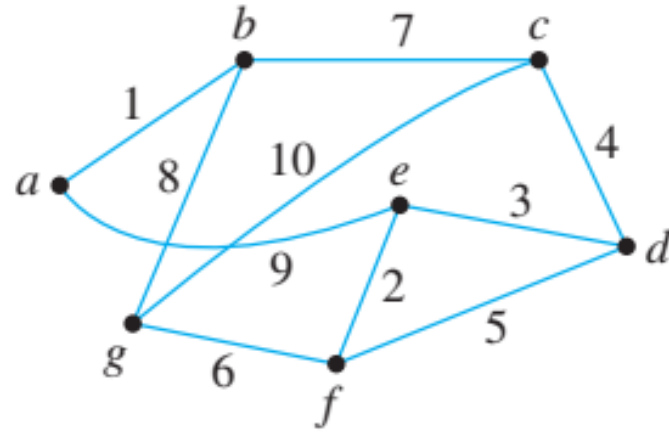
1. Consider the tree shown at right with root a .
- What is the level of n ?
 - What is the level of a ?
 - What is the height of this rooted tree?
 - What are the children of n ?
 - What is the parent of g ?
 - What are the siblings of j ?
 - What are the descendants of f ?



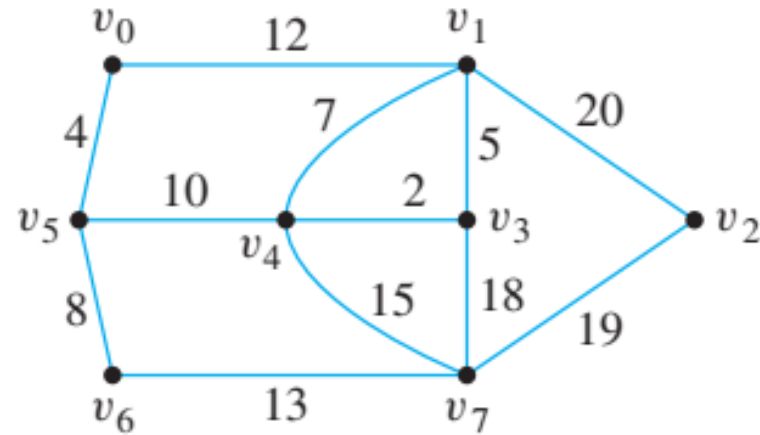
Exercise Set 10.7

Use Kruskal's algorithm to find a minimum spanning tree for each of the graphs in 5 and 6. Indicate the order in which edges are added to form each tree.

5.



6.

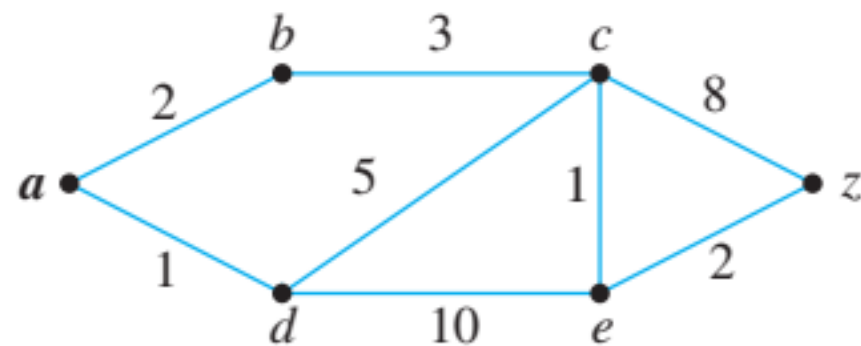


Use Prim's algorithm starting with vertex a or v_0 to find a minimum spanning tree for each of the graphs in 7 and 8. Indicate the order in which edges are added to form each tree.

7. The graph of exercise 5. 8. The graph of exercise 6.

Use Dijkstra's algorithm to find the shortest path from a to z for each of the graphs in 13–16. In each case make tables similar to Table 10.7.1 to show the action of the algorithm.

13.



14.

